

Sebastopol College

Econometrics for Managers

Target group: 3rd year Management students

By: Gashaw M. (PhD)

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Brief contents

Chapter1: Introduction

Chapter2: Simple Linear Regression

Chapter3: Multiple Linear Regression

Chapter4: Binary Dependent Models

Chapter 5: Introduction toTime Series Models

* STATA LAB(Optional)

Recommended Texts

1. Damodar N. Gujarati, *Basic Econometrics*, Fourth Edition. New York: McGraw-Hill, 2003. ISBN 0-07-233542-4.
2. Jeffrey M. Wooldridge, *Introductory Econometrics: A Modern Approach*, Fourth Edition. South-Western Cengage Learning, 2009.

Purpose and Learning Objectives

The purpose of the course is to help students **learn to use econometric methods in empirical research.**

The course focuses exclusively on econometric models and methods designed for analysis of ***empirical data***.

Econometrics is an integral part of training in modern business. Econometric applications provide organizations with a potent set of tools to unlock the power of information and in effective decision making.

Modern managers use econometric applications intoo their business to establish healthy economic strategy, optimize solutions & outperform competition.

- Upon successful completion of the course, students should have acquired the methodological, operational and expositional skills required **to *design* and *conduct* a small-scale econometric investigation.**

Why Study Econometrics?

Though Econometrics is fundamental for economic measurement, its application extends far beyond the discipline of economics.

It is a set of research tools employed in all business disciplines.

It is also used by social scientists.

This breadth of interest in econometrics arises b/c of its appropriateness of quantifying relationships.

So, through this training you become a person who better understands how you can quantify quantitative and qualitative variables using cross sectional data.

- In short the importance of econometrics can be summarize as:
 - i. Describing economic reality
 - ii. Testing hypotheses about any theory, and
 - iii. Forecasting future economic activity.

What Is Econometrics About?

- It is about to:
 - Quantifying relationships b/n variables
 - Answering the “how much” question
 - Interpreting the reality
 - Forecasting events
- It all begins with a theory from your field of study—about how important variables are related to one another.
- Let's, in economics, see r/ships b/n consumption expenditure and current income: $\text{Consumption} = f(\text{Income})$

This is a general economic model of consumption.

- But, for most economic decision, it is not enough to know that certain economic variables are interrelated.
- We must be able to say “how much” a change in one affects another.

Definition and scope of econometrics

- Econometrics deals with the quantitative study of economic relations.
- It gives a quantitative answer/assign magnitude to r/ships.
- It is a tool to interpret reality in the light of the theory, using statistical techniques.
- The **economic theories** we learn in various economics courses suggest many r/ships among economic variables. For instance the r/ship b/n:
 - Quantity demand and price
 - Consumption and income
 - Investment and rate of interest, etc

Cont'd

- Econometricians used different ways of wordings to define econometrics.
- But, the concepts are the same: means ‘metrics’ to imply “measurement”.
- Hence, **Econometrics deals with the measurement of economic relationships.**
- In other words, econometrics may be considered as the **integration of theory, mathematics, and statistics** for the purpose of providing numerical values for the parameters of economic relationships and verifying economic theories.
- However, it is completely **distinct** from each one of these three branches of science.

- a) Economic theory states a qualitative relationship b/n variables using Ceteris Paribus assumption.

Ex.1. Consumption depends up on current income (X_t) & previous income (X_{t-1}) of an individual other things being constant.

- While econometrics provides relationships with numerical statements.
- b) Mathematical Economics: it explains the economic theory in the equation or mathematical form.

E.g. 2: $Y_i = \beta_1 + \beta_2 X_i$ where X_i is income of unit i

Then Mathematical Economics explain the exact r/ship b/n the dependent (Y) and independent variable (X)

Is income the only determinant of consumption?

- c) Mathematical Statistics: these are based up on the probability theory which are developed on the basis of controlled experiment. They are applied in economic relation ships.
- d) Econometrics model: mathematical model is a deterministic form. Consumption can also be affected by wealth, windfall income/loss, tastes/preferences, policies, etc

So by putting all these in the error term, “U”, we can specify the econometric model as

- $Y_i = \beta_1 + \beta_2 X_i + U_i$ where “U” is the error term, which represents all those other factors (everything else) affecting consumption expenditure

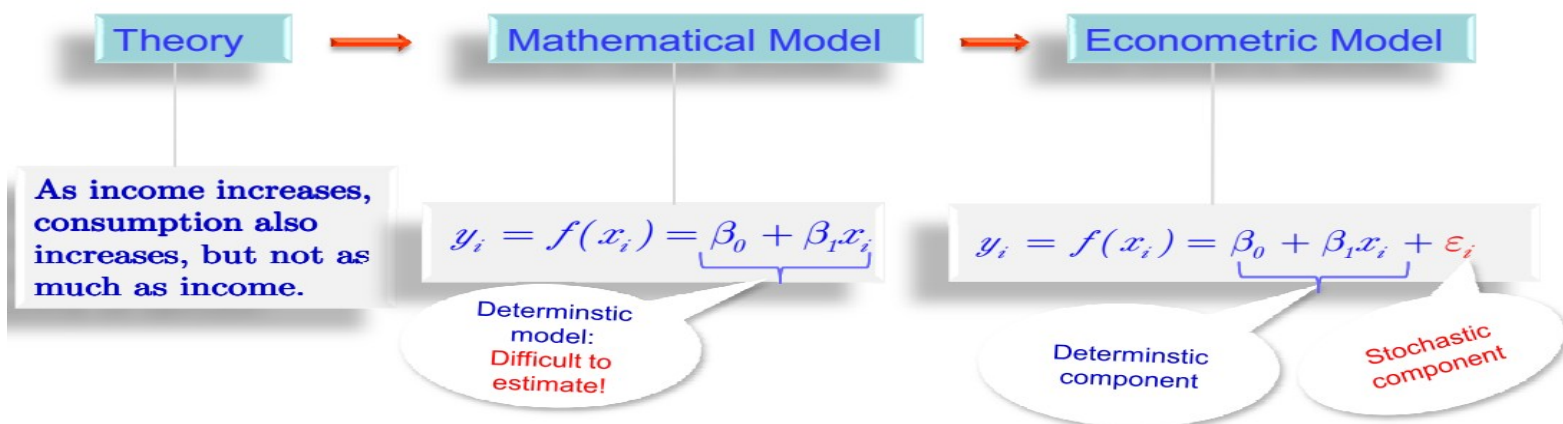
Econometrics model

- To specify of the econometrics model, we will do the following
 - selecting variables
 - determine the theoretical values
 - specify the mathematical model
 - specify the econometric model

N.B.: Due to the randomness of the dependent variable, the error term is included in the linear mathematical model.

Cont'd

- ❑ Econometrics is the application of statistical and mathematical techniques to the analysis of economic data with a purpose of verifying economic theories.
- ❑ It is the unified study of **economic theory, mathematics, & statistics** to measure economic r/ships b/n economic variables.



- ❑ Thus, Econometrics is about how to best estimate economic parameters given the data we have.

Reasons to include the error term

1. Omission of variables. For example, income is not the only determinants of consumption.
2. Measurement error in collecting data. — We may use poor proxy variables.
3. The functional form may not be correct.
4. Vagueness of theory
5. Unavailability of data
6. Erratic (random = unpredictable) human behaviour
7. Error of aggregation:- the sum of the parts may be different from the whole
8. Principle of Parsimony
9. Core variables versus peripheral variables

Methodology of Econometric Research

Procedures in econometrics research

- Start with economic theory **at least take common sense**
- Mathematical formulation
- Econometrics model
- Data collection
- Analysis (estimating econometric model)
- Evaluation of estimates (hypothesis testing & inference)
- Forecasting (Application)

Goals of Econometrics

- Testing of Economic theories /**Analysis**/:- **the explanation power of economic theory is tested by econometrics = verification of Economic theories**
- Estimation of the coefficients of Economic relationship
/Policy making/
$$Y_i = \alpha + \beta_1 I + \beta_2 Ex + \beta_3 PI + \beta_4 PEx + U_i$$
- Forecasting the future values of economic magnitudes.
/Forecasting/

Successful econometric application should include some combination of all the three goals.

$$\hat{Y}_i = -261.09 + 0.2453 X_i$$

Econometric Data

- a. **Cross-sectional data:** collected across sample units in a particular time period on one/more variables.

Ex. Number of children registered for schooling in all K.G. Schools of D/Tabor in 2016 E.C. by sex, age, religion etc.

- b. **Time series data:** data collected over discrete intervals of time.
The key feature of it is that the same economic quantity is recorded at a regular time interval.

Ex. Annual real gross domestic product (GDP)

Year	GDP
2001	11347.2
2002	11553.0
2003	11840.7
2004	12263.8
2005	12638.4
2006	12976.2
2007	13254.1
2008	13312.2

c. **Panel data**:- also known as “longitudinal” data, has observations on individual micro-units (cross-sections) who are followed over time.

Ex. If consumption expenditure of a sample of population from D/Tabor for some basket is taken in 1985, in 1990 & 1996.

Or, Data on two rice farms

<i>FIRM</i>	<i>YEAR</i>	<i>PROD</i>	<i>AREA</i>	<i>LABOR</i>	<i>FERT</i>
1	1990	7.87	2.50	160	207.5
1	1991	7.18	2.50	138	295.5
1	1992	8.92	2.50	140	362.5
1	1993	7.31	2.50	127	338.0
1	1994	7.54	2.50	145	337.5
1	1995	4.51	2.50	123	207.2
1	1996	4.37	2.25	123	345.0
1	1997	7.27	2.15	87	222.8
2	1990	10.35	3.80	184	303.5
2	1991	10.21	3.80	151	206.0
2	1992	13.29	3.80	185	374.5
2	1993	18.58	3.80	262	421.0
2	1994	17.07	3.80	174	595.7
2	1995	16.61	4.25	244	234.8
2	1996	12.28	4.25	159	479.0
2	1997	14.20	3.75	133	170.0

d. **Polled data**:- These are data of both time series & cross sectional data.

N.B.: Dummy Variable: qualitative data, ex. sex, religion, race, profession etc.

Estimation of the Econometric Model

Various methods to estimate
the parameters:

OLS

ILS

2SLS

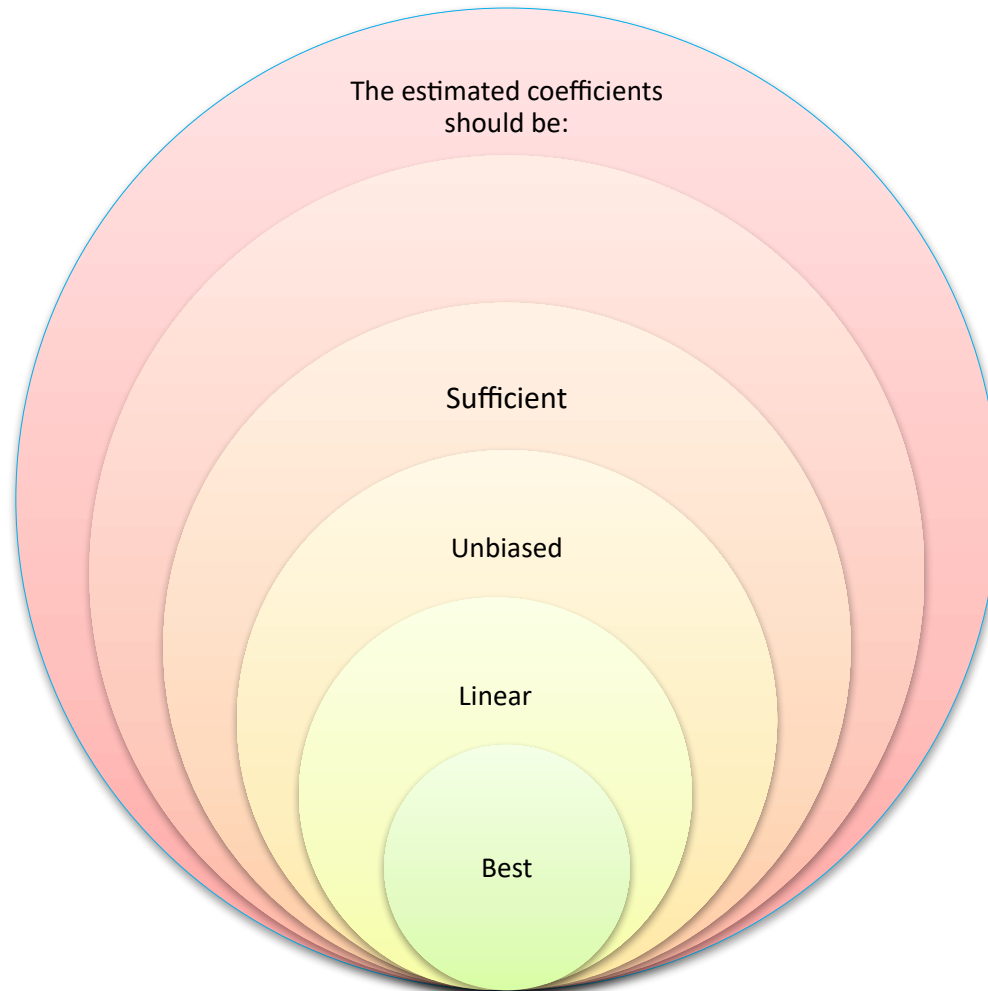
3SLS

MLM

MM

GMM

Properties: Estimated Coefficients



Evaluations of Estimates

- At this stage we are evaluating the reliability of the results whether they are theoretically meaningful & statistically significant.

Includes:

- i. Economic interpretation /Economic a priori criterion.
- ii. Statistical interpretation of the results - statistical analysis on the basis of R^2 , t-test, F- test, s.d.
- iii. Test of Econometric criterion.

Chapter Two

Simple Linear Regression Model

Objectives

- ✓ Explain the simplest type of regression analysis in which the relationship b/n the variables is approximated by a straight line.
- ✓ Discuss & explain each of the assumptions of the simple linear regression model.

Regression, Causation, and Correlation

- ❑ Causation means that a change in one variable is the direct consequence of a change in another variable.
- ❑ Regression is a method for studying the r/ship b/n a response variable Y & a covariate X. Regression does not necessarily imply causation. **Causation** comes from **theory rather than statistics.**
- ❑ Correlation does not imply causation. There are many associations b/n variables that are not causal. E.g. the association that those who had been in a hospital are less healthy does not imply going to the hospital causes a person to be less healthy.
- ❑ Correlation measures the strength of linear association, while regression measures the magnitude too.
- ❑ Economic theory suggests many r/ships b/n economic variables.

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- But, by **how much** a change in one variable affect the other?

Can we forecast or predict the corresponding value of another?

- A regression model answers the 'how much' question.
- Regression analysis is a statistical method that uses data to explore relationships between variables.
- In regression, we have stochastic dependent variable and non-stochastic independent variable , while in correlation variables involved are stochastic.

E.g. In **income-expenditure experiment**, the weekly consumption expenditure has a **probability** values.

- If X_i = weekly household income then the conditional probability density is $f(Y/X=x)$.

Cont'd

- Simple linear regression analysis examines the r/ship b/n a y-variable and one x-variable.
- Let us assume that this relationship is linear,

$$Y_i = \beta_1 + \beta_2 X_i \dots\dots\dots(2.1)$$

where $i=1, 2, 3, \dots, N$

- In practice, due to the randomness of Y, the r/ship (2.1) will not hold exactly. Each Y_i will be only explained up to an error term U_i , so that

$$Y_i = \beta_1 + \beta_2 X_i + U_i \dots\dots\dots(2.2)$$

where Y refers to as the:

- dependent variable, or LHS, or Explained, or Regresand Variable

- X refers to as the:

- Independent Variable, or RHS, or Explanatory, or Regressor, or Covariate, or Control Variable
- U_i is an error term (is also a random variable).
- β_1 & β_2 are parameters

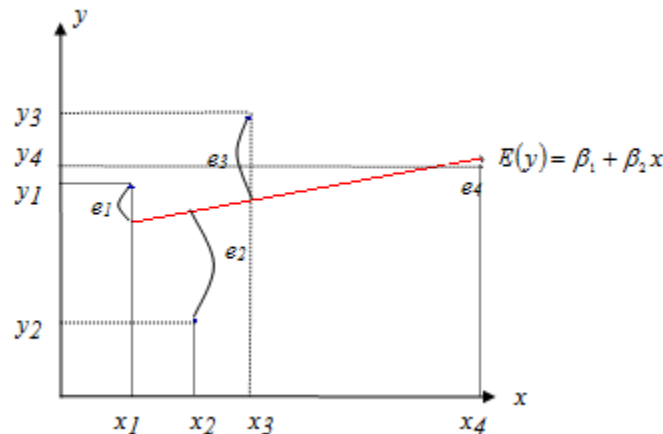
- The randomness of the Y variable results from that of the error term.

Population and sample regression function

- Regression is estimation or prediction of the average value of a dependent variable on the basis of the fixed values of other variables.
- When the variable X is deterministic, from (2.2) it follows that:

$$E(Y_i) = \beta_1 + \beta_2 X_i \dots\dots\dots(2.3)$$

where $\beta_1 + \beta_2 X_i$ is the population regression function.



- ❑ The best linear approximation of y from x and a constant is obtained by minimizing the sum of squared residuals.
- ❑ All fitted values are on a straight line, the regression line:
 $E(y|x) = \beta_1 + \beta_2 X_i$, where β_1 and β_2 are unknown but fixed parameters and are called the **regression coefficients**.

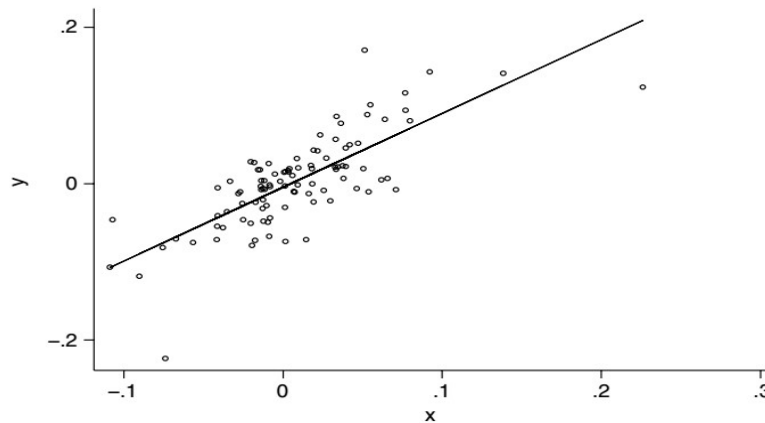


Fig: Simple linear regression: fitted line & observation points

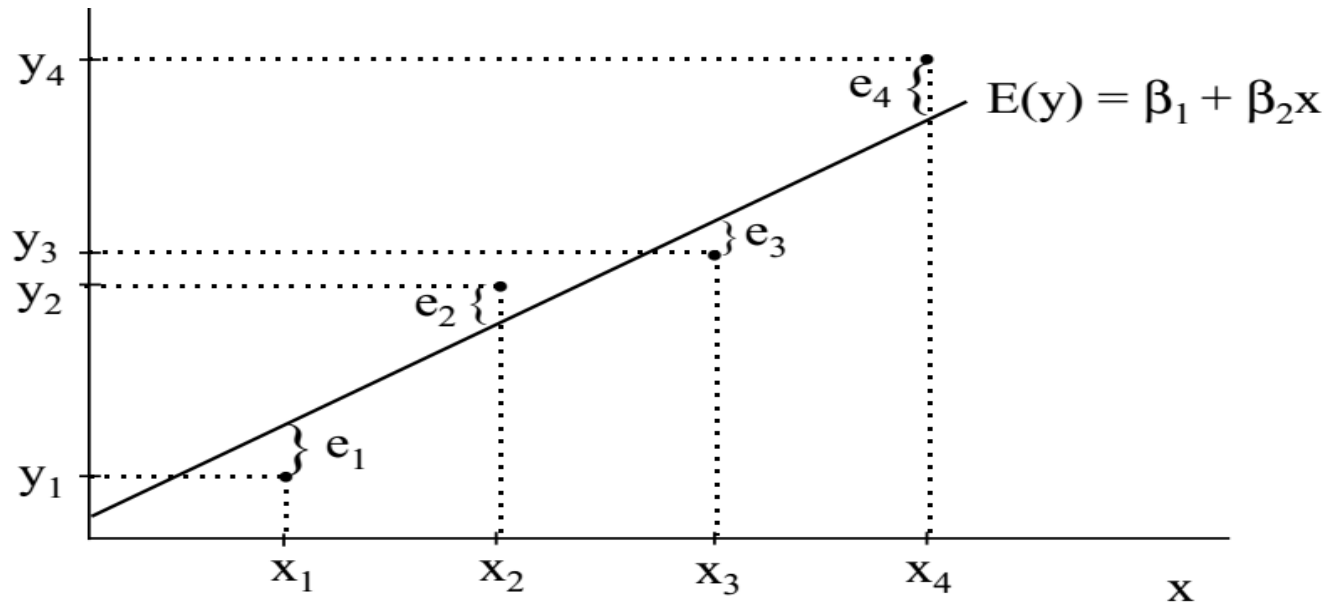


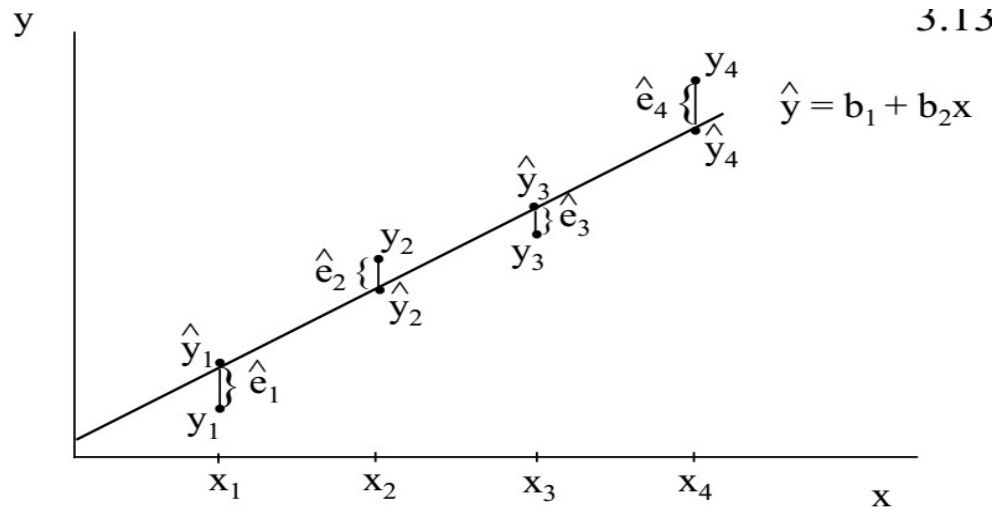
Fig: The fitted line and the residual

Sample regression function

- The sample counter part of the PRF is the sample regression function (SRF).
- The estimates are estimators of the true population parameters:

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 \hat{X}_i \text{ ----- (2.4)}$$

where $\hat{\beta}_1$ & $\hat{\beta}_2$ are estimates of β_1 & β_2 respectively.



- ❑ Thus the population and sample regression lines are:

$$E(Y_i/X_i) = \beta_1 + \beta_2 X_i; \quad \text{and} \quad \hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 \hat{x}_i \text{ respectively.}$$

- ❑ Are results for the SRF the same as that of the PRF? The answer is no; due to sampling fluctuation
- ❑ The task is to make sure the estimates are as close as the population parameters

Assumptions of SLR

1. Linearity: (Linearity in variables and in parameters)

$$y = \beta_1 + \beta_2 x + u$$

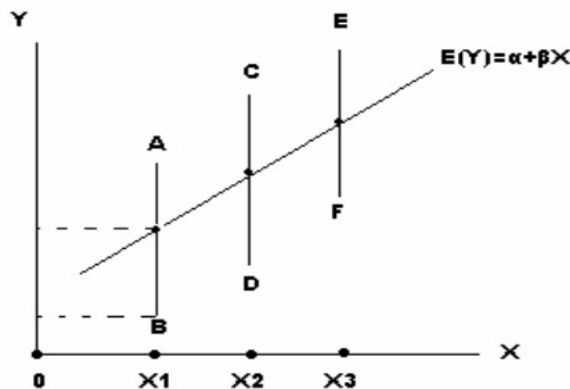
2. The mean value of y , for each value of x , is given by the *linear regression*: $E(y | x) = \beta_1 + \beta_2 x$

3. The expected value of the error term is zero:

$$E(u/x) = E(u) = 0, \text{ w/h implies } E(y/x) = b_0 + b_1 x$$

4. U is a random real variable

5. Homoscedasticity(constant variance of the error term):



Proof :

$$\text{Var}(u | x) = E(u^2 | x) - [E(u | x)]^2$$

$$E(u | x) = 0,$$

$$\text{so } \sigma^2 = E(u^2 | x) = E(u^2) = \text{Var}(u) = \text{Var}(y)$$

6. The random terms of d/t observations (U_i, U_j) are independent
(No autocovariance)

$$\text{Cov}(u_i, u_j) = E(u_i u_j) = 0$$

7. U must normally distributed with mean zero and variance (δ^2)

$$u \sim N(0, \sigma_u^2)$$

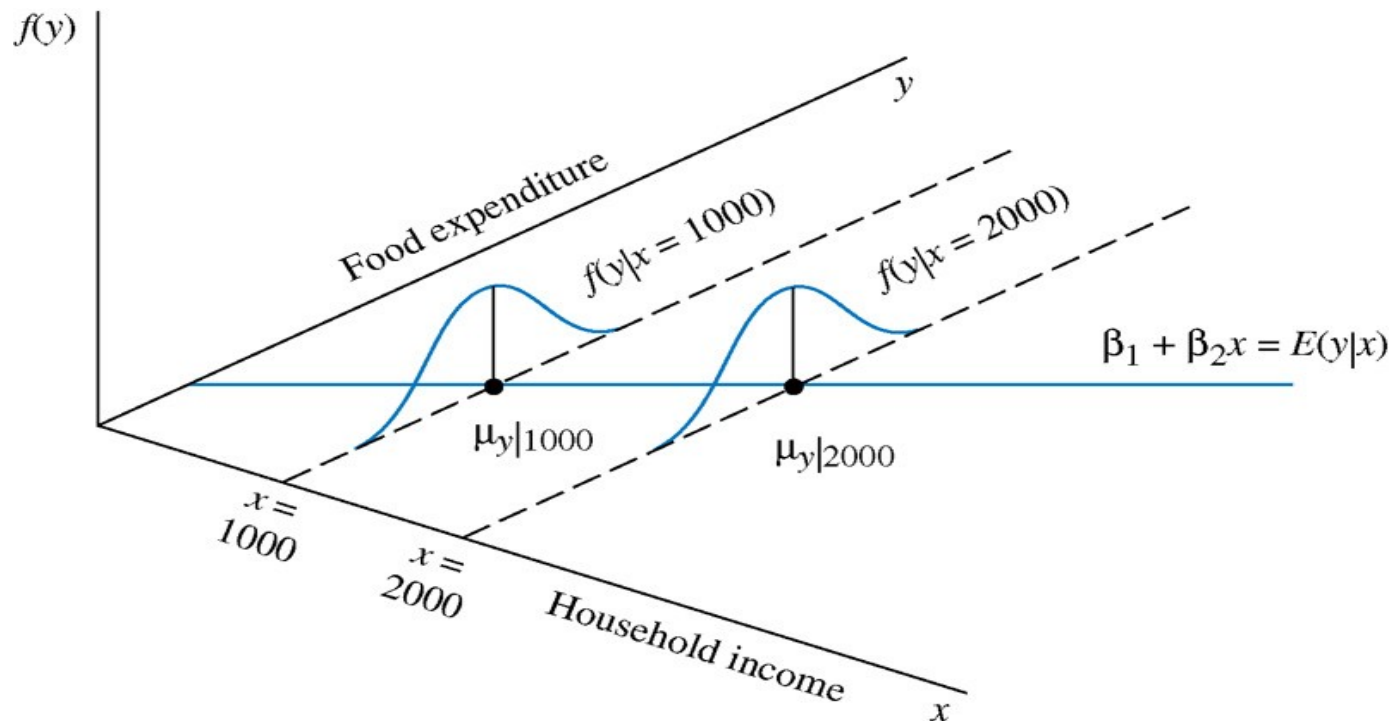
8. Zero covariance

$$\begin{aligned} \text{cov}(u_i, X_i) &= E [u_i - E(u_i)][X_i - E(X_i)] \\ &= E [u_i(X_i - E(X_i))] \text{ since } E(u_i) = 0 \\ &= E (u_i X_i) - E(X_i)E(u_i) \\ &= E(u_i X_i) \text{ since } E(u_i) = 0 \\ &= 0 \text{ by assumption} \end{aligned}$$

9. X need to vary in the sample but must be fixed in repeated samples.

10. Number of observations must be greater than the parameters estimated
11. The variable x is not random, and must take at least two different values

The assumptions can be summarized as:



Methods of Estimation

- Given the above assumptions, the SLR model is defined as

- $$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i \quad \text{-----} \quad (2.2^*)$$

where, $E(\varepsilon_i/X_i) = 0$ and $V(\varepsilon_i/X_i) = \sigma^2$

- The predicted/fitted values or line is

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 \hat{X}_i \quad \text{-----} \quad (2.4^*)$$

- And the residuals are defined to be the difference b/n the actual value of Y in any observation and the fitted value given by the regression line is denoted by ε_i :

$$\text{i.e. } \varepsilon_i = Y_i - \hat{Y}_i = \hat{Y}_i - (\hat{\beta}_1 + \hat{\beta}_2 \hat{X}_i) \quad \text{-----} \quad (2.5)$$

Cont'd

- The residual sums of squares or RSS, which measures how well the line fits the data, is defined by

$$RSS = \sum_{i=1}^n \hat{\epsilon}_i^2$$

- The least squares estimates are the values β_1 and β_2 that

$$\text{minimizes } RSS = \sum_{i=1}^n \hat{\epsilon}_i^2$$

- Thus, the least squares estimates are given by

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sum_{i=1}^n (X_i - \bar{X}_n)^2}$$

$$\hat{\beta}_1 = \bar{Y}_n - \hat{\beta}_2 \bar{X}_n$$

- How to calculate good estimates of these values algebraically?

The ordinary least square (OLS) method

- Depending on the nature of the data and functional form, there are different methods of estimating the parameters.
- One of the most widely used methods is ordinary least squares (OLS) estimation.
- Let the true r/ship model is:

$$Y_i = \alpha + \beta X_i + U_i \text{-----} (2.6)$$

where α & β are the true parameters.

- And the corresponding estimated r/ship model is:

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta} X_i + \hat{U}_i = \hat{\alpha} + \hat{\beta} X_i \text{-----} (2.4 **)$$

$\hat{Y}_i$ is the best linear approximation of Y_i .

- The first step is defining a residual for each observation.
- Then the CLS (OLS) method chooses estimators $\hat{\alpha}$ & $\hat{\beta}$ so as to minimize the sum the squares of the d/c e b/n the actual and fitted values of Y:

$$\min \sum (e_i)^2 = \sum (Y_i - \hat{Y}_i)^2 \text{ --- (2.7)}$$

where $\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$

That is $\sum e_i^2 = \sum (Y_i - \hat{\alpha} - \hat{\beta}X_i)^2 \text{ --- (2.7 *)}$

To find the values of $\hat{\alpha}$ & $\hat{\beta}$ that minimizes this sum, we use the concept of calculus called [derivatives](#).

Thus,

$$1. \frac{\partial \sum e_i^2}{\partial \hat{\alpha}} = -2 \sum (Y_i - \hat{\alpha} - \hat{\beta}X_i) = 0 \dots \dots \dots (2.8)$$

Rearranging this expression we will get: $\boxed{\sum Y_i = n\alpha + \hat{\beta}\sum X_i} \dots\dots(2.9)$

If you divide (2.9) by ‘n’ and rearrange, we get

$$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X} \dots\dots\dots(2.10)$$

$$2. \frac{\partial \sum e_i^2}{\partial \hat{\beta}} = -2 \sum X_i (Y_i - \hat{\alpha} - \hat{\beta}X) = 0 \dots\dots\dots(2.11)$$

If we rearrange equation (2.11) we obtain;

$$\boxed{\sum Y_i X_i = \hat{\alpha} \sum X_i + \hat{\beta} \sum X_i^2} \text{-----}(2.12)$$

Equation (2.9) and (2.12) are called the Normal Equations.

- By substituting the values of $\hat{\alpha}$

$$\begin{aligned}\sum Y_i X_i &= \sum X_i (\bar{Y} - \hat{\beta} \bar{X}) + \hat{\beta} \sum X_i^2 \\ &= \bar{Y} \sum X_i - \hat{\beta} \bar{X} \sum X_i + \hat{\beta} \sum X_i^2\end{aligned}$$

$$\sum Y_i X_i - \bar{Y} \sum X_i = \hat{\beta} (\sum X_i^2 - \bar{X} \sum X_i)$$

$$\sum XY - n \bar{X} \bar{Y} = \hat{\beta} (\sum X_i^2 - n \bar{X}^2)$$

$$\hat{\beta} = \frac{\sum XY - n \bar{X} \bar{Y}}{\sum X_i^2 - n \bar{X}^2} \text{ ----- (2.13)}$$

- Equation (2.13) can be rewritten in [deviation](#) form as follows:

$$\begin{aligned}\Sigma(X - \bar{X})(Y - \bar{Y}) &= \Sigma(XY - X\bar{Y} - \bar{X}Y + \bar{X}\bar{Y}) \\ &= \Sigma XY - \bar{Y} \Sigma X - \bar{X} \Sigma Y + n \bar{X} \bar{Y} \\ &= \Sigma XY - n \bar{Y} \bar{X} - n \bar{X} \bar{Y} + n \bar{X} \bar{Y} \\ &= \Sigma XY - n \bar{X} \bar{Y} \text{ ----- (2.14)}\end{aligned}$$

And;

$$\Sigma(X - \bar{X})^2 = \Sigma X^2 - n \bar{X}^2 \text{ ----- (2.15)}$$

- Thus,

$$\hat{\beta} = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{\Sigma(X - \bar{X})^2}$$

Now, denoting $(X_i - \bar{X})$ as x_i , and $(Y_i - \bar{Y})$ as y_i we get;

$$\hat{\beta} = \frac{\Sigma x_i y_i}{\Sigma x_i^2} \text{----- (2.16)}$$

The estimated residuals are

$$\hat{u}_i = y_i - \hat{\alpha} - \hat{\beta}x_i$$

The two normal equations show that these residuals satisfy the equations

$$\sum \hat{u}_i = 0 \quad \text{and} \quad \sum x_i \hat{u}_i = 0$$

The residual sum of squares (to be denoted by RSS) is given by

$$\begin{aligned} \text{RSS} &= \sum (y_i - \hat{\alpha} - \hat{\beta}x_i)^2 \\ &= \sum [y_i - \bar{y} - \hat{\beta}(x_i - \bar{x})]^2 \\ &= \sum (y_i - \bar{y})^2 + \hat{\beta}^2 \sum (x_i - \bar{x})^2 - 2\hat{\beta} \sum (y_i - \bar{y})(x_i - \bar{x}) \\ &= S_{yy} + \hat{\beta}^2 S_{xx} - 2\hat{\beta} S_{xy} \quad \text{----- (2.17)} \end{aligned}$$

But $\hat{\beta} = S_{xy}/S_{xx}$. Hence we have

$$RSS = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = S_{yy} - \hat{\beta}S_{xy}$$

Thus
$$\begin{array}{ccccc} \text{TSS} & = & \text{ESS} & + & \text{RSS} \\ \text{(total)} & & \text{(explained)} & & \text{(residual)} \end{array}$$

Variance & Standard errors of OLS estimates

□ In statistics the precision of an estimate is measured by its standard error (se).

Then the variances and covariance of α and β with out proof are:

$$var(\hat{\alpha}) = \delta^2 \left[\frac{\sum X_i^2}{n \sum x_i^2} \right] \Rightarrow se(\hat{\alpha}) = \delta \left[\sqrt{\frac{\sum X_i^2}{n \sum x_i^2}} \right] \text{----- (2.18)}$$

And

$$var(\hat{\beta}) = \frac{\delta^2}{\sum x_i^2} \Rightarrow se(\hat{\beta}) = \frac{\delta}{\sqrt{\sum x_i^2}} \text{----- (2.19)}$$

Where $\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n-2}$ and $\sum \hat{u}_i^2 = \sum y_i^2 - \hat{\beta}_2^2 \sum x_i^2$

$$\text{And} \quad cov(\hat{\alpha}, \hat{\beta}) = -\bar{X} var(\hat{\beta}) = -\bar{X} \left[\frac{\sigma^2}{\sum x_i^2} \right]$$

Gauss-Markov Theorem: Properties of OLS Estimators

- Given the assumptions of the classical linear regression model, the least-squares estimates possess some optimum properties contained in the well-known Gauss–Markov theorem.
 - Estimators, say α and β are said to be **best linear unbiased estimators (BLUE)** of the true α and β .
 - Under the above assumptions, the estimators α and β have the smallest variance of all linear and unbiased estimators of α and β .
1. It is **linear**, that is, a linear function of a random variable, such as the dependent variable Y in the regression model.
 2. It is **unbiased**, that is, its average/expected value, $E(\hat{\beta})$, is equal to the true value, β .
 3. It has **minimum variance** in the class of all such linear unbiased estimators; an unbiased estimator with the least variance is known as an efficient estimator.

The Coefficient of Determination: r^2 (R^2)

- How do we think about how well our sample regression line fits our sample data?
- The coefficient of determination r^2 (two-variable case) or R^2 (multiple regression) is a summary measure that tells how well the sample regression line fits the data.
- It is taken as a measure of “***Goodness -of- Fit***”; i.e. it shows the percentage of the total variation of the dependent variable that can be explained by the independent variable X.
- Given $Y_i = \beta_1 + \beta_2 X_i + U_i$, r^2 -measures the proportion of the variation in Y explained by X_i

- Then r^2 is defined as

$$r^2 = \frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum(Y_i - \bar{Y})^2} = \frac{\text{ESS}}{\text{TSS}} \text{-----} (2.20)$$

or, alternatively, as

$$\begin{aligned} r^2 &= 1 - \frac{\sum \hat{u}_i^2}{\sum(Y_i - \bar{Y})^2} \\ &= 1 - \frac{\text{RSS}}{\text{TSS}} \end{aligned} \text{-----} (2.20^*)$$

Two properties of r^2 may be noted:

1. It is a nonnegative quantity. (Why?)
2. Its limits are $0 \leq r^2 \leq 1$. An r^2 of 1 means a perfect fit, that is, $\hat{Y}_i = Y_i$ for each i . On the other hand, an r^2 of zero means that there is no relationship between the regressand and the regressor whatsoever (i.e., $\hat{\beta}_2 = 0$). In

- Or, r^2 can be quickly calculated from

$$\begin{aligned}
 r^2 &= \frac{ESS}{TSS} \\
 &= \frac{\sum \hat{y}_i^2}{\sum y_i^2} \\
 &= \frac{\hat{\beta}_2^2 \sum x_i^2}{\sum y_i^2} \\
 &= \hat{\beta}_2^2 \left(\frac{\sum x_i^2}{\sum y_i^2} \right) \text{----- (2.20**) }
 \end{aligned}$$

- The value of r^2 falls b/n zero and one. i.e. $0 \leq r^2 \leq 1$.
- **Interpretation:**
Suppose $r^2 = 0.9$, this means that the regression line gives a good fit to the observed data since this line explains 90% of the total variation of the Y value around their mean.

A numerical example

- Weekly family consumption expenditure Y and weekly family income X

Y	70	65	90	95	110	115	120	140	155	150
X	80	100	120	140	160	180	200	220	240	260

- Given the data, obtain the estimates of the regression coefficients of the model:

$$Y_i = \beta_1 + \beta_2 X_i + U_i$$

- From these raw data, the following calculations are obtained.
(See next slide).

$$\hat{\beta}_1 = 24.4545 \quad \text{var}(\hat{\beta}_1) = 41.1370 \quad \text{and} \quad \text{se}(\hat{\beta}_1) = 6.4138$$

$$\hat{\beta}_2 = 0.5091 \quad \text{var}(\hat{\beta}_2) = 0.0013 \quad \text{and} \quad \text{se}(\hat{\beta}_2) = 0.0357$$

$$\text{cov}(\hat{\beta}_1, \hat{\beta}_2) = -0.2172 \quad \hat{\sigma}^2 = 42.1591$$

$$r^2 = 0.9621 \quad r = 0.9809 \quad \text{df} = 8$$

The estimated regression line therefore is

$$\hat{Y}_i = 24.4545 + 0.5091X_i$$

- Check for the results by showing the necessary procedures.

Estimation procedure

Y_i	X_i	$Y_i X_i$	X_i^2	$\begin{matrix} x_i = \\ X_i - \bar{X} \end{matrix}$	$\begin{matrix} y_i = \\ Y_i - \bar{Y} \end{matrix}$	x_i^2	$x_i y_i$	$\hat{Y}_i$	$\begin{matrix} \hat{u}_i = \\ Y_i - \hat{Y}_i \end{matrix}$	$\hat{Y}_i \hat{u}_i$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
70	80	5600	6400	-90	-41	8100	3690	65.1818	4.8181	314.0524
65	100	6500	10000	-70	-46	4900	3220	75.3636	-10.3636	-781.0382
90	120	10800	14400	-50	-21	2500	1050	85.5454	4.4545	381.0620
95	140	13300	19600	-30	-16	900	480	95.7272	-0.7272	-69.6128
110	160	17600	25600	-10	-1	100	10	105.9090	4.0909	433.2631
115	180	20700	32400	10	4	100	40	116.0909	-1.0909	-126.6434
120	200	24000	40000	30	9	900	270	125.2727	-6.2727	-792.0708
140	220	30800	48400	50	29	2500	1450	136.4545	3.5454	483.7858
155	240	37200	57600	70	44	4900	3080	145.6363	8.3636	1226.4073
150	260	39000	67600	90	39	8100	3510	156.8181	-6.8181	-1069.2014
Sum 1110	1700	205500	322000	0	0	33000	16800	1109.9995 ≈ 1110.0	0	0.0040 ≈ 0.0
Mean 111	170	nc	nc	0	0	nc	nc	110	0	0
$\hat{\beta}_2 = \frac{\sum x_i y_i}{\sum x_i^2} = 16,800 / 33,000 = 0.5091$ $\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X} = 111 - 0.5091(170) = 24.4545$										

Interpretation:

- The value of $\hat{\beta}_2 = 0.5091$, *which* measures the slope of the line, shows that, within the sample range of X b/n \$80 & \$260 per week, as X *increases, say, by \$1, the estimated* increase in the mean or average weekly consumption expenditure amounts to about 51 cents.
- The value of $\hat{\beta}_1 = 24.4545$, which is the intercept of the line, indicates the average level of weekly consumption expenditure when weekly income is zero. However, this is a mechanical interpretation. In regression analysis such literal interpretation of the intercept term may not be always meaningful, although in the present example it can be argued that a family without any income might maintain some minimum level of consumption expenditure either by borrowing or dis-saving.

Interpretation...

- The value of r^2 of 0.9621 means that about 96 percent of the variation in the weekly consumption expenditure is explained by income. Since r^2 can at most be 1, the observed r^2 suggests that the sample regression line fits the data very well.
- The coefficient of correlation of 0.9809 shows that the two variables, consumption expenditure and income, are highly positively correlated.

Exercise

Sales(Y_i)	Temperature (X_i)
7	1
8	2
9	3
10	4
11	5
12	6
13	7

Given the above observations estimate the parameters

Numerical Example 2

Sales(Y) depends on advertisement(X)

$$\text{Model: } Y_i = \beta_0 + \beta_1 X_i + U_i$$

Obsn (i)	(Y)	(X)	y=Y-Ybar	x=X-Xbar	x*y	x*x	Yhat
1	36	56.7	-16.583333	-9.43333	156.4361	88.98778	34.3847
2	48	63.9	-4.5833333	-2.23333	10.23611	4.987778	48.27482
3	45	62.7	-7.5833333	-3.43333	26.03611	11.78778	45.9598
4	40	59.7	-12.583333	-6.43333	80.95278	41.38778	40.17225
5	30	55.9	-22.583333	-10.2333	231.1028	104.7211	32.84135
6	56	68.7	3.41666667	2.566667	8.769444	6.587778	57.5349
7	63	69.2	10.4166667	3.066667	31.94444	9.404444	58.4995
8	53	65.5	0.41666667	-0.63333	-0.26389	0.401111	51.36152
9	61	69.4	8.41666667	3.266667	27.49444	10.67111	58.88533
10	68	73.4	15.4166667	7.266667	112.0278	52.80444	66.60207
11	66	74.1	13.4166667	7.966667	106.8861	63.46778	67.9525
12	65	74.4	12.4166667	8.266667	102.6444	68.33778	68.53125
Sum	631	793.6	-2.842E-14	1.14E-13	894.2667	463.5467	631
Mean	52.58333	66.133333 3	-2.368E-15	9.47E-15			

CCC

$\hat{\beta}_1 =$						1.92918369					
$\hat{\beta}_1 =$	1.92918369					-75.000014					
$\hat{\beta}_0 =$	-75.000014										
Fitted equation is	Yhat =	-75	+	1.929184	X						

Regression Output:	
Constant	-75
Std Err of Y Est	2.443604
R Squared	0.966546
No. of Observations	12
Degrees of Freedom	10
X Coefficient(s)	1.92918369
Std Err of Coef.	0.11349692
t-value	16.9976738

CCC

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	1725.204664	1725.205	288.9209	1.04583E-08
Residual	10	59.71200311	5.9712		
Total	11	1784.916667	162.265152	Root MSE =	2.4436
<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	
Intercept	-75.00001438	7.539004122	-9.94827	1.67E-06	-91.79796528
X Variable 1	1.929183685	0.113496923	16.99767	1.05E-08	1.67629673

Numerical example 3: Output & labor-hours work data

<i>Observation</i>	<i>x</i>	<i>y</i>	<i>x</i> ²	<i>y</i> ²	<i>xy</i>
1	10	11	100	121	110
2	7	10	49	100	70
3	10	12	100	144	120
4	5	6	25	36	30
5	8	10	64	100	80
6	8	7	64	49	56
7	6	9	36	81	54
8	7	10	49	100	70
9	9	11	81	121	99
10	<u>10</u>	<u>10</u>	<u>100</u>	<u>100</u>	<u>100</u>
Total	80	96	668	952	789

$$\bar{x} = \frac{80}{10} = 8$$

$$\bar{y} = \frac{96}{10} = 9.6$$

Obs	Y _i	X _i	Y _i ²	X _i ²	YX	Y_hat t	U_hat	U_hat_ sqr
1	11	10	121	100	110	11.1	-0.1	0.01
2	10	7	100	49	70	8.85	1.15	1.3225
3	12	10	144	100	120	11.1	0.9	0.809999
4	6	5	36	25	30	7.35	-1.35	1.8225
5	10	8	100	64	80	9.6	0.4	0.16
6	7	8	49	64	56	9.6	-2.6	6.76
7	9	6	81	36	54	8.1	0.9	0.809999
8	10	7	100	49	70	8.85	1.15	1.3225
9	11	9	121	81	99	10.35	0.65	0.422499
10	10	10	100	100	100	11.1	-1.1	1.21
Sum	96	80	952	668	789		0	14.65
Mean	9.6	8	95.2	66.8	78.9			

$$\bar{x} = \frac{80}{10} = 8$$

$$\bar{y} = \frac{96}{10} = 9.6$$

$$S_{xx} = 668 - 10(8)^2 = 668 - 640 = 28$$

$$S_{xy} = 789 - 10(8)(9.6) = 789 - 768 = 21$$

$$S_{yy} = 952 - 10(9.6)^2 = 952 - 921.6 = 30.4$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{21}{28} = 0.75$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 9.6 - 0.75(0.8) = 3.6$$

Hence the regression of y on x is

$$y = 3.6 + 0.75x$$

Interpretation of a Regression Equation

- N.B.: The interpretation is conditional on the equation being correctly specified.
- When Y and X are variables with straightforward natural units (not logarithms or other functions),
 - the slope coefficient indicates that, as X increases by one unit, Y increases by $\beta^{\wedge}$ units.
 - the constant term indicates the predicted level of Y when X is 0.

Numerical Example: Interpretation

reg EARNINGS S

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
				Adj R-squared	=	0.1710
				Root MSE	=	13.126

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

$$\text{Earnings} = -13.93 + 2.46S$$

- ✓ So the slope coefficient implies that hourly earnings increase by \$2.46 for each extra year of schooling
- ✓ Literally, the constant indicates that an individual with no years of education would have to pay \$13.93 per hour to be allowed to work

Exercise 2

- Given sales and advertisement relation as

Observation	Y (sales)	X (Ads)
1	40	100
2	50	200
3	50	300
4	70	400
5	65	500
6	65	600
7	70	700

Constant	36.42857
R ²	0.845497
No. of Obs	7
df	5
Coefficient	0.058929
S.e. Coef.	0.011266

Exercise

Obs	Consumption	Income			
i	Y	X	Y2	X2	XY
1	52.3	36.4	2735.29	1324.96	1903.72
2	78.44	46.8	6152.834	2190.24	3670.992
3	88.76	57.2	7878.338	3271.84	5077.072
4	54.08	67.6	2924.646	4569.76	3655.808
5	111.4	74.3	12409.96	5520.49	8277.02
6	105.2	86.5	11067.04	7482.25	9099.8
7	45.73	91.3	2091.233	8335.69	4175.149
8	122.35	102.8	14969.52	10567.84	12577.58
9	142.24	114.5	20232.22	13110.25	16286.48
10	86.22	120.9	7433.888	14616.81	10424
11	174.5	135	30450.25	18225	23557.5
12	185.2	144	34299.04	20736	26668.8
13	111.8	156	12499.24	24336	17440.8
14	214.6	173.7	46053.16	30171.69	37276.02
15	144.6	182	20909.16	33124	26317.2
16	174.36	199.2	30401.41	39680.64	34732.51
17	215.4	208	46397.16	43264	44803.2
18	286.24	217.8	81933.34	47436.84	62343.07
19	188.56	223.2	35554.87	49818.24	42086.59
20	237.2	234	56263.84	54756	55504.8
21	181.8	251	33051.24	63001	45631.8
22	373	260	139129	67600	96980
23	191.6	289.5	36710.56	83810.25	55468.2
24	247.12	296.4	61068.29	87852.96	73246.37
25	269.6	312	72684.16	97344	84115.2
sum	4082.3	4080.1	825299.7	832146.8	801319.7

Analysis of variance (ANOVA) in regression

- We already know that for simple CLRM $Y_i = \beta_0 + \beta_1 X_i + u_i$:

$$TSS = ESS + RSS \quad \text{And for } d_i = X_i - \bar{X},$$

$$\Rightarrow \sum y_i^2 = \sum \hat{y}_i^2 + \sum \hat{u}_i^2 = \hat{\beta}_1^2 \sum d_i^2 + \sum \hat{u}_i^2$$

In tabular form, this ANOVA in regression looks like:

Source of variation	Sum of squares(SS)	df	Mean SS(=SS/df)
Due to regression (ESS)	$\sum \hat{y}_i^2 = \hat{\beta}_1^2 \sum d_i^2$	1	$\hat{\beta}_1^2 \sum d_i^2$
Due to residuals (RSS)	$\sum \hat{u}_i^2$	$n - 2$	$\frac{\sum \hat{u}_i^2}{n-2} = \hat{\sigma}^2$
Total (TSS)	$\sum y_i^2$	$n - 1$	$\frac{\sum y_i^2}{n-1}$

Properties of the Ordinary Least Squares (OLS) Estimators

- The ideal or optimum properties that the OLS estimates possess be summarized by the Gauss-Markov Theorem
- i.e. the OLS estimators are BLUE

Best (Minimum Variance)

Linear(linear function of the a random variable, such as, Y

Unbiased(its expected value = true population parameter)

Estimator

- Proposition: $\hat{\alpha}$ & $\hat{\beta}$ are linear in Y.

Proof:
$$\hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2} = \frac{\sum x_i (Y - \bar{Y})}{\sum x_i^2} = \frac{\sum x_i Y - \bar{Y} \sum x_i}{\sum x_i^2},$$

$$(\text{but } \sum x_i = \sum (X - \bar{X}) = \sum X - n\bar{X} = n\bar{X} - n\bar{X} = 0)$$

$$\Rightarrow \hat{\beta} = \frac{\sum x_i Y}{\sum x_i^2}; \quad \text{Now, let } \frac{x_i}{\sum x_i^2} = K_i \quad (i = 1, 2, \dots, n)$$

$$\therefore \hat{\beta} = \sum K_i Y$$

w/c shows that $\hat{\beta}$ is a linear estimator b/c it is a linear function of Y

- **Unbiasedness:**

Proposition: $\hat{\alpha}$ & $\hat{\beta}$ are the unbiased estimators of the true parameters α & β

- if $\hat{\theta}$ is the unbiased estimator of θ then bias = 0
i.e. $E(\hat{\theta}) - \theta = 0 \Rightarrow E(\hat{\theta}) = \theta$

We know that $\hat{\beta} = \frac{\sum k_i Y_i}{\sum k_i^2} = \frac{\sum k_i (\alpha + \beta X_i + U_i)}{\sum k_i^2}$

$$= \frac{\alpha \sum k_i + \beta \sum k_i X_i + \sum k_i U_i}{\sum k_i^2},$$

$$\text{but } \sum k_i = 0 \quad \text{and} \quad \sum k_i X_i = 1$$

$$\sum k_i = \frac{\sum x_i}{\sum x_i^2} = \frac{\sum (X - \bar{X})}{\sum x_i^2} = \frac{\sum X - n\bar{X}}{\sum x_i^2} = \frac{n\bar{X} - n\bar{X}}{\sum x_i^2} = 0$$

$$\Rightarrow \sum k_i = 0$$

$$\begin{aligned}\sum k_i X_i &= \frac{\sum x_i X_i}{\sum x_i^2} = \frac{\sum (X - \bar{X}) X_i}{\sum x_i^2} \\ &= \frac{\sum X^2 - \bar{X} \sum X}{\sum X^2 - n \bar{X}^2} = \frac{\sum X^2 - n \bar{X}^2}{\sum X^2 - n \bar{X}^2} = 1\end{aligned}$$

$$\Rightarrow \sum k_i X_i = 1$$

$$\hat{\beta} = \beta + \sum k_i u_i \Rightarrow \hat{\beta} - \beta = \sum k_i u_i$$

$$E(\hat{\beta}) = E(\beta) + \sum k_i E(u_i), \text{ Since } k_i \text{ are fixed}$$

$$E(\hat{\beta}) = \beta, \quad \text{since } E(u_i) = 0$$

Therefore, $\hat{\beta}$ is unbiased estimator of β .

Minimum variance of

- Now, we have to establish that out of the class of linear and unbiased estimators of β , possess the smallest sampling variances.

Variance of $\hat{\beta}$

$$\text{var}(\beta) = E(\hat{\beta} - E(\hat{\beta}))^2 = E(\hat{\beta} - \beta)^2$$

$$\text{var}(\hat{\beta}) = E(\sum k_i u_i)^2$$

$$= E[k_1^2 u_1^2 + k_2^2 u_2^2 + \dots + k_n^2 u_n^2 + 2k_1 k_2 u_1 u_2 + \dots + 2k_{n-1} k_n u_{n-1} u_n]$$

$$= E[k_1^2 u_1^2 + k_2^2 u_2^2 + \dots + k_n^2 u_n^2] + E[2k_1 k_2 u_1 u_2 + \dots + 2k_{n-1} k_n u_{n-1} u_n]$$

$$= E(\sum k_i^2 u_i^2) + E(\sum k_i k_j u_i u_j) \quad i \neq j$$

$$= \sum k_i^2 E(u_i^2) + 2 \sum k_i k_j E(u_i u_j) = \sigma^2 \sum k_i^2 \quad (\text{Since } E(u_i u_j) = 0)$$

$$\sum k_i = \frac{\sum x_i}{\sum x_i^2}, \text{ and therefore, } \sum k_i^2 = \frac{\sum x_i^2}{(\sum x_i^2)^2} = \frac{1}{\sum x_i^2}$$

$$\therefore \text{var}(\hat{\beta}) = \sigma^2 \sum k_i^2 = \frac{\sigma^2}{\sum x_i^2}$$

Now, let us check whether this variance of OLS estimator do possess minimum variance property compared to the variance other estimator of the true β , other than $\hat{\beta}$, say β^* .

Lets first show minimum variance of $\hat{\beta}$

1. Minimum variance of $\hat{\beta}$

Suppose: β^* an alternative linear & unbiased estimator of β and

$$\text{Let } \beta^* = \sum w_i Y_i$$

$$\text{where, } w_i \neq k_i ; \text{ but: } w_i = k_i + c_i$$

$$\beta^* = \sum w_i (\alpha + \beta X_i + u_i) \text{ Since } Y_i = \alpha + \beta X_i + U_i$$

$$= \alpha \sum w_i + \beta \sum w_i X_i + \sum w_i u_i$$

$$\therefore E(\beta^*) = \alpha \sum w_i + \beta \sum w_i X_i, \text{ since } E(u_i) = 0$$

Since β^* is assumed to be an unbiased estimator, then for β^* to be an unbiased estimator of β , there must be true that $\sum w_i = 1$ and $\sum w_i X_i = \sum X_i / n$ in the above equation.

$$\text{But, } w_i = k_i + c_i$$

$$\Sigma w_i = \Sigma(k_i + c_i) = \Sigma k_i + \Sigma c_i$$

Therefore, $\Sigma c_i = 0$ since $\Sigma k_i = \Sigma w_i = 0$

Again $\Sigma w_i X_i = \Sigma(k_i + c_i)X_i = \Sigma k_i X_i + \Sigma c_i X_i$

Since $\Sigma w_i X_i = 1$ and $\Sigma k_i X_i = 1 \Rightarrow \Sigma c_i X_i = 0$.

From these values we can drive $\Sigma c_i x_i = 0$, where $x_i = X_i - \bar{X}$

Thus, from the above calculations we can summarize the following results.

$$\sum w_i = 0, \quad \sum w_i x_i = 1, \quad \sum c_i = 0, \quad \sum c_i X_i = 0$$

To prove whether $\hat{\beta}$ has minimum variance or not let's compute $\text{var}(\beta^*)$ to compare with $\text{var}(\hat{\beta})$.

$$\text{var}(\beta^*) = \text{var}(\sum w_i Y_i)$$

$$= \sum w_i^2 \text{var}(Y_i)$$

$$\therefore \text{var}(\beta^*) = \sigma^2 \sum w_i^2 \quad \text{since } \text{Var}(Y_i) = \sigma^2$$

$$\text{But, } \sum w_i^2 = \sum (k_i + c_i)^2 = \sum k_i^2 + 2\sum k_i c_i + \sum c_i^2$$

$$\Rightarrow \sum w_i^2 = \sum k_i^2 + \sum c_i^2 \quad \text{Since } \sum k_i c_i = \frac{\sum c_i x_i}{\sum x_i^2} = 0$$

Therefore, $\text{var}(\beta^*) = \sigma^2(\Sigma k_i^2 + \Sigma c_i^2) \Rightarrow \sigma^2 \Sigma k_i^2 + \sigma^2 \Sigma c_i^2$

$$\boxed{\text{var}(\beta^*) = \text{var}(\hat{\beta}) + \sigma^2 \Sigma c_i^2}$$

Given that c_i is an arbitrary constant, $\sigma^2 \Sigma c_i^2$ is a positive i.e it is greater than zero.

Thus $\text{var}(\beta^*) > \text{var}(\hat{\beta})$. This proves that $\hat{\beta}$ possesses minimum variance property.

In the similar way we can prove that the least square estimate of the constant intercept ($\hat{\alpha}$) possesses minimum variance.

The variance of the random variable (U_i)

- the unbiased estimate of σ^2 which is $\hat{\sigma}^2$ computed from the sample value of the disturbance term e_i from the expression:

$$\hat{\sigma}_u^2 = \frac{\sum e_i^2}{n-2} \quad \text{i.e., } E(\hat{\sigma}^2) = E\left(\frac{\sum e_i^2}{n-2}\right) = \sigma^2$$

Proof:

$$\left. \begin{array}{l} Y_i = \hat{\alpha} + \hat{\beta}X_i + e_i \\ \hat{Y} = \hat{\alpha} + \hat{\beta}x \end{array} \right\}$$

$$\Rightarrow Y = \hat{Y} + e_i$$

$$\Sigma Y_i = \Sigma y_i + \Sigma e_i$$

$$\Sigma Y_i = \Sigma \hat{Y}_i \quad \text{since } (\Sigma e_i) = 0$$

Dividing both sides the above by 'n' will give us

$$\frac{\Sigma Y}{n} = \frac{\Sigma \hat{Y}_i}{n} \rightarrow \bar{Y} = \bar{\hat{Y}}.$$

$$\left. \begin{array}{l} Y = \hat{Y} + e \\ \bar{Y} = \bar{\hat{Y}} \end{array} \right\}$$

$$\Rightarrow (Y - \bar{Y}) = (\hat{Y} - \bar{\hat{Y}}) + e$$

$$\Rightarrow y_i = \hat{y}_i + e \dots\dots\dots$$

$$e_i = y_i - \hat{y}_i.$$

Now, we have to express y_i and $\hat{y}_i$ in other expression as derived below.

$$\text{From: } \left. \begin{array}{l} Y_i = \alpha + \beta X_i + U_i \\ \bar{Y} = \alpha + \beta \bar{X} + \bar{U} \end{array} \right\}$$

We get, by subtraction

$$y_i = (Y_i - \bar{Y}) = \beta_i (X_i - \bar{X}) + (U_i - \bar{U}) = \beta x_i + (U - \bar{U})$$

$$\Rightarrow y_i = \beta x + (U - \bar{U})$$

Note that we assumed earlier that , $E(u)=0$,

Similarly: From;

$$\left. \begin{aligned}\hat{Y} &= \hat{\alpha} + \hat{\beta}x \\ \bar{Y} &= \hat{\alpha} + \hat{\beta}\bar{x}\end{aligned}\right\}$$

We get, by subtraction

$$\hat{Y} - \bar{Y} = \hat{\beta}(x - \bar{x})$$

$$\Rightarrow \hat{y} = \hat{\beta}x$$

$$e_i = \beta x_i + (u_i - \bar{u}) - \hat{\beta}x_i$$

$$= (u_i - \bar{u}) - (\hat{\beta}_i - \beta)x_i$$

The summation over the n sample values of the squares of the residuals over the 'n' samples yields:

$$\Sigma e_i^2 = \Sigma [(u_i - \bar{u}) - (\hat{\beta} - \beta)x_i]^2$$

$$= \Sigma [(u_i - \bar{u})^2 + (\hat{\beta} - \beta)^2 x_i^2 - 2(u_i - \bar{u})(\hat{\beta} - \beta)x_i]$$

$$= \Sigma (u_i - \bar{u})^2 + (\hat{\beta} - \beta)^2 \Sigma x_i^2 - 2[(\hat{\beta} - \beta)\Sigma x_i(u_i - \bar{u})]$$

Taking expected values we have:

$$E(\sum e_i^2) = E[\sum(u_i - \bar{u})^2] + E[(\hat{\beta} - \beta)^2 \sum x_i^2] - 2E[(\hat{\beta} - \beta) \sum x_i (u_i - \bar{u})]$$

The right hand side terms of this equation may be rearranged as follows

$$\begin{aligned} \text{a. } E[\sum(u - \bar{u})^2] &= E(\sum u_i^2 - \bar{u} \sum u_i) \\ &= E\left(\sum u_i^2 - \frac{(\sum u_i)^2}{n}\right) \\ &= \sum E(u_i^2) - \frac{1}{n} E(\sum u)^2 \\ &= n\sigma^2 - \frac{1}{n} E(u_1 + u_2 + \dots + u_i)^2 \text{ since } E(u_i^2) = \sigma_u^2 \\ &= n\sigma^2 - \frac{1}{n} E(\sum u_i^2 + 2\sum u_i u_j) \\ &= n\sigma^2 - \frac{1}{n} (\sum E(u_i^2) + 2\sum u_i u_j) \quad i \neq j \\ &= n\sigma^2 - \frac{1}{n} n\sigma_u^2 - \frac{2}{n} \sum E(u_i u_j) \\ &= n\sigma_u^2 - \sigma_u^2 \text{ (given } E(u_i u_j) = 0) \\ &= \sigma_u^2 (n - 1) \end{aligned}$$

$$\text{b. } E[(\hat{\beta} - \beta)^2 \Sigma x_i^2] = \Sigma x_i^2 \cdot E(\hat{\beta} - \beta)^2$$

Given that the X's are fixed in all samples and we know that

$$E(\hat{\beta} - \beta)^2 = \text{var}(\hat{\beta}) = \sigma_u^2 \frac{1}{\Sigma x^2}$$

$$\text{Hence } \Sigma x_i^2 \cdot E(\hat{\beta} - \beta)^2 = \Sigma x_i^2 \cdot \sigma_u^2 \frac{1}{\Sigma x^2}$$

$$\boxed{\Sigma x_i^2 \cdot E(\hat{\beta} - \beta)^2 = \sigma_u^2}$$

$$\text{c. } -2E[(\hat{\beta} - \beta)\Sigma x_i(u_i - \bar{u})] = -2E[(\hat{\beta} - \beta)(\Sigma x_i u_i - \bar{u}\Sigma x_i)]$$

$$= -2E[(\hat{\beta} - \beta)(\Sigma x_i u_i)], \text{ since } \Sigma x_i = 0$$

$$-2 E[(\hat{\beta} - \beta) \sum x_i (u_i - \bar{u})] = -2 E[(\sum k_i u_i)(\sum x_i u_i)]$$

$$= -2 E \left[\left(\frac{\sum x_i u_i}{\sum x_i^2} \right) (\sum x_i u_i) \right] \quad , \text{since } k_i = \frac{x_i}{\sum x_i^2}$$

$$= -2 E \left[\frac{(\sum x_i u_i)^2}{\sum x_i^2} \right]$$

$$= -2 E \left[\frac{\sum x_i^2 u_i^2 + 2 \sum x_i x_j u_i u_j}{\sum x_i^2} \right]$$

$$= -2 \left[\frac{\sum x_i^2 E(u_i^2)}{\sum x_i^2} + \frac{2 \sum (x_i x_j) E(u_i u_j)}{\sum x_i^2} i \neq j \right]$$

$$= -2 \frac{\sum x_i^2 E(u_i^2)}{\sum x_i^2} \quad (\text{given } E(u_i u_j) = 0)$$

$$\boxed{= -2 E(u_i^2) = -2 \sigma^2}$$

Consequently, the above equation can be written as

$$E(\sum e_i^2) = (n-1)\sigma_u^2 + \sigma^2 - 2\sigma_u^2 = (n-2)\sigma_u^2.$$

From which we get

$$E\left(\frac{\sum e_i^2}{n-2}\right) = E(\hat{\sigma}_u^2) = \sigma_u^2$$

$$\text{Since } \hat{\sigma}_u^2 = \frac{\sum e_i^2}{n-2}$$

Thus, $\hat{\sigma}^2 = \frac{\sum e_i^2}{n-2}$ is unbiased estimate of the true variance of the error term (σ^2).

Interval Estimation and Hypothesis Testing

- An estimator is likely to be different from the true parameter due to sampling fluctuations.
- But in a repeated sampling, the expected value is equal to the true value: $E(\hat{\beta}) = \beta$
- The reliability of the estimator in representing the true parameter is measured by its *standard error*.
- The true parameter can be at some distance to the left or right side of the point estimator in terms of *standard errors*.
- We construct interval around the point two or three standard errors on either side. This is the essence of *interval estimation*.

Confidence Interval

- Assume that we want to see how close $\hat{\beta}$ is to β .
- Consider two positive numbers δ and α where the probability that random interval $(\hat{\beta} - \delta, \hat{\beta} + \delta)$ contains the true β is $1 - \alpha$ and $0 < \alpha < 1$.
- Symbolically:

$$\Pr[\hat{\beta} - \delta \leq \beta \leq \hat{\beta} + \delta] = 1 - \alpha$$

- This interval is called *confidence interval*.
- $1 - \alpha$ is called the confidence coefficient.
- α is called the level of significance (probability of committing *Type I error*)

-
- The interval has a probability of $1-\alpha$ to include the true β within its limit.
 - It does not say that the probability of β to lie between the interval is $1-\alpha$. Because β is a fixed number, though not known, it is either in the interval or not.
 - Rather, we say the probability of constructing an interval containing β is $1-\alpha$.
 - As $\hat{\beta}$ is random and as the interval is based on it, then the interval is also random.
 - Since the interval is random, we understand it in a long run, repeated sampling. That is, it is on average in repeated sampling that the interval contains β is $(1-\alpha)$ of the cases.

- Once we have the sample and get a specific number for $\hat{\beta}$, then the interval is no longer random. It is fixed.
- This time, we cannot say that the probability of the interval containing β is $(1-\alpha)$. Rather, the probability is either 1 or 0.
- If, for example, we have a 95% confidence interval such as:

$(0.427 \leq \beta \leq 0.59)$, for our hypothetical consumption-income example

we cannot say that the probability that this interval contains β is 95%. Rather, it is either 1 or 0.

- with the normality assumption for u_i , *the OLS estimators are themselves normally distributed* with means and variances given therein. Therefore, for example, the variable

$$Z = \frac{\hat{\beta}_2 - \beta_2}{\text{se}(\hat{\beta}_2)}$$

$$= \frac{(\hat{\beta}_2 - \beta_2)\sqrt{\sum x_i^2}}{\sigma}$$

It therefore seems that we can use the normal distribution to make probabilistic statements about β s provided the true population variance σ^2 is *known where* a normally distributed variable with mean μ and variance σ^2 is that the area under the normal curve b/n $\mu \pm \sigma$ is about 68%, that b/n the limits $\mu \pm 2\sigma$ is about 95%, and that b/n $\mu \pm 3\sigma$ is about 99.7%.

Confidence Interval for $\hat{\beta}_1$

- The estimator $\hat{\beta}_1$ is normally distributed.

$$Z = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{(\hat{\beta}_1 - \beta_1)\sqrt{\sum x_i^2 - n\bar{x}^2}}{\sigma}$$

- But σ is rarely known. If we replace σ by $\hat{\sigma}$, we have:

$$t = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} = \frac{(\hat{\beta}_1 - \beta_1)\sqrt{\sum x_i^2 - n\bar{x}^2}}{\hat{\sigma}}$$

- The t variable defined follows a t -distribution with $n-2$ degrees of freedom. Hence,

$$Pr\left(-t_{\frac{\alpha}{2}} \leq t \leq t_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

Confidence Interval for $\hat{\beta}_1$

- The confidence interval for $\hat{\beta}_1$ is thus:

$$Pr\left(-t_{\frac{\alpha}{2}} \leq \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} \leq t_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\Rightarrow Pr\left[\hat{\beta}_1 - t_{\frac{\alpha}{2}} se(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{\frac{\alpha}{2}} se(\hat{\beta}_1)\right] = 1 - \alpha$$

- Once we have interval like (0.4268, 0.5914)) the 95% confidence interval interpretation is that, in the long run, in 95 out of 100 cases, intervals like this will contain the true β .
- The probability that the specified fixed interval includes the true β is 1 or 0; not 95%.
- The larger is the standard error, the larger is the width of the confidence interval.

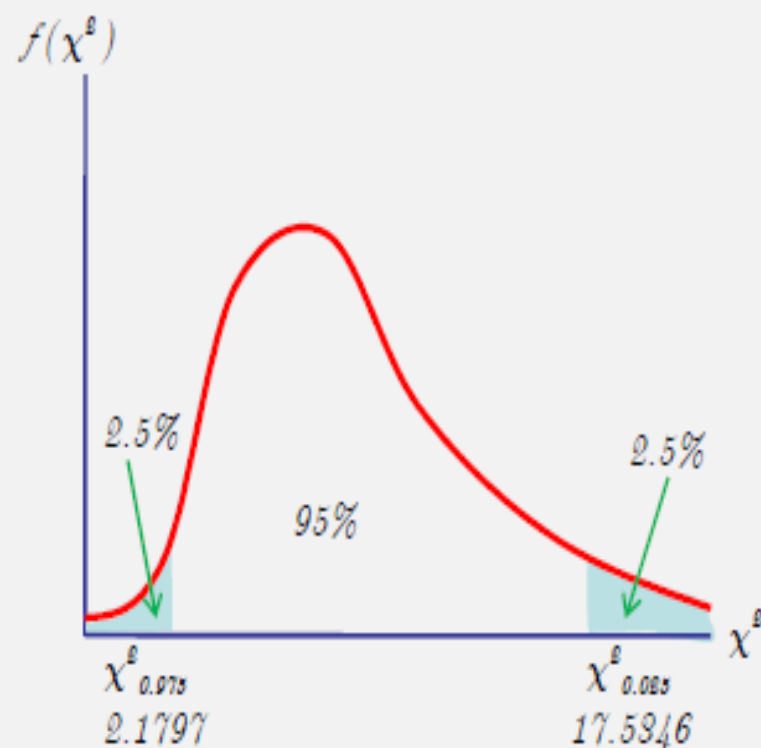
Confidence Interval for σ^2

- We notice that $\chi^2 = (n-2) \frac{\hat{\sigma}^2}{\sigma^2}$ follows χ^2 -distribution.
- Then,

$$\Pr \left[\chi^2_{1-\frac{\alpha}{2}} \leq \chi^2 \leq \chi^2_{\frac{\alpha}{2}} \right] = 1 - \alpha$$

$$\Rightarrow \Pr \left[(n-2) \frac{\hat{\sigma}^2}{\chi^2_{\frac{\alpha}{2}}} \leq \sigma^2 \leq (n-2) \frac{\hat{\sigma}^2}{\chi^2_{1-\frac{\alpha}{2}}} \right] = 1 - \alpha$$

gives the $100(1-\alpha)\%$ confidence interval for σ^2 .



Interval Estimation and Hypothesis Testing: HT

- HT is a process(set of procedure) of checking whether or not a given finding is sufficiently close to some predetermined(stated) value
- Evidently, there are two types of Hypotheses
- the stated or *null hypothesis* denoted by H_0
- the alternative (maintained)hypothesis denoted by H_1

- There are two mutually complementary approaches to HT
 - Test of Significance Approach(TSA)
 - Confidence Interval Approach(CIA)

Hypothesis Testing: TSA

- The decision to accept or reject H_0 is made on the basis of the value of the test statistic
- The t-test and F -test are most common to test coefficients and standard errors

Procedures:

1. Find coefficient estimates and its standard error
Example $\hat{\beta}_1$ and $se(\hat{\beta}_1)$
2. Set up your hypothesis about the true parameter

$$H_0 : \beta_1 = \beta_1^*$$

$$H_1 : \beta_1 (><) \beta_1^*$$

Or

$$H_1 : \beta_1 \neq \beta_1^*$$

3. Under normality assumption construct the t-statistic(calculated t-value)

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{se(\hat{\beta}_1)}$$

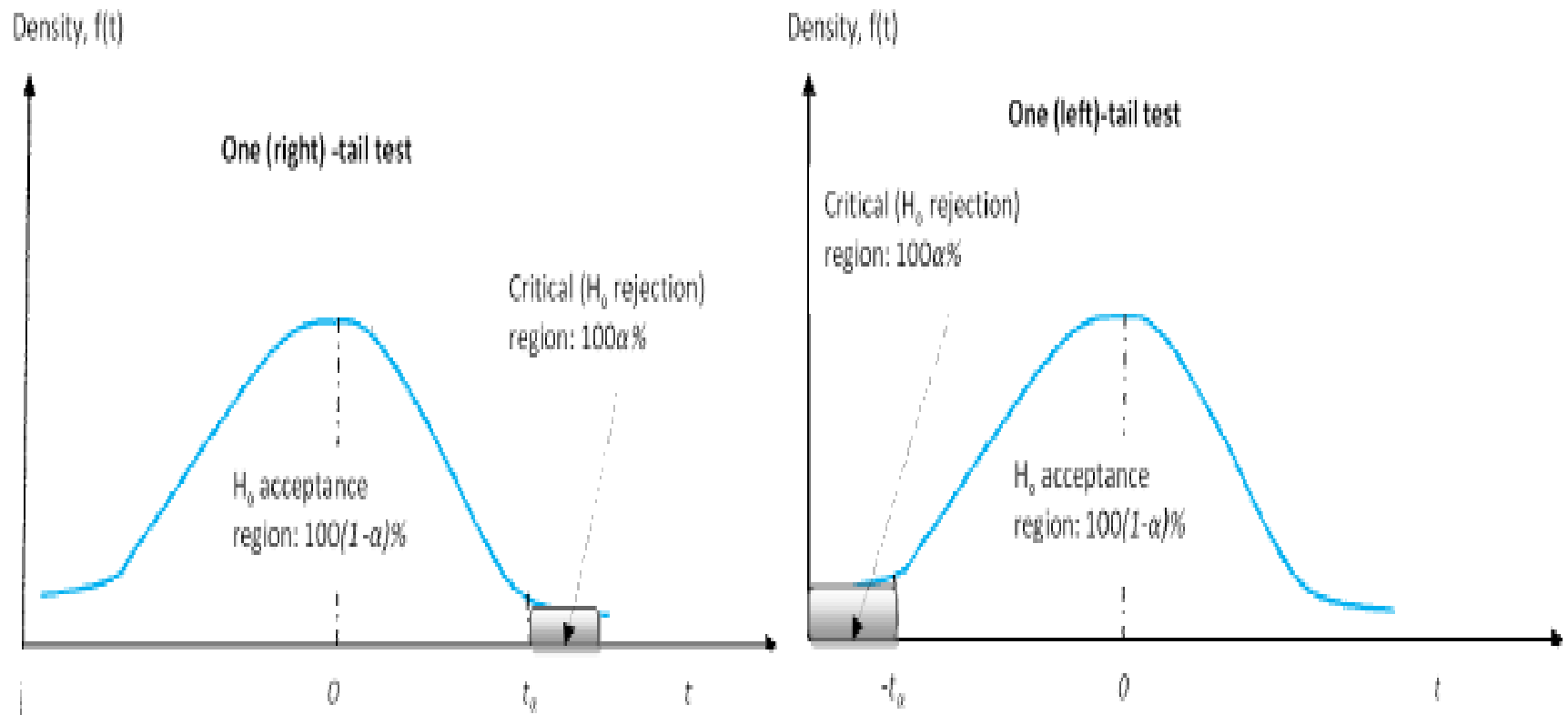
4. Based on the calculated $df (= n - k)$, predetermined α and type of H_1 (one tail or two tail), read the critical(tabulated) t that is

$-t_{\alpha/2}$ and $t_{\alpha/2}$ for two tailed alternative hypothesis
 $-t_{\alpha}$ or t_{α} for one tailed alternative hypothesis

5. Decision:

- If $|t| > t_{\text{tab}}$ i.e., if $|t|$ becomes larger & larger, we reject $H_0: \beta_1 = \beta_1^*$. It lies in the rejection region.

CC...



- **Example:** A regression equation from the income consumption data of 10 countries has given the following output.

$$mpc = \hat{\beta}_1 = 0.5091, se(\hat{\beta}_1) = 0.0357$$

$$H_0 : \beta_1^* = 0.3$$

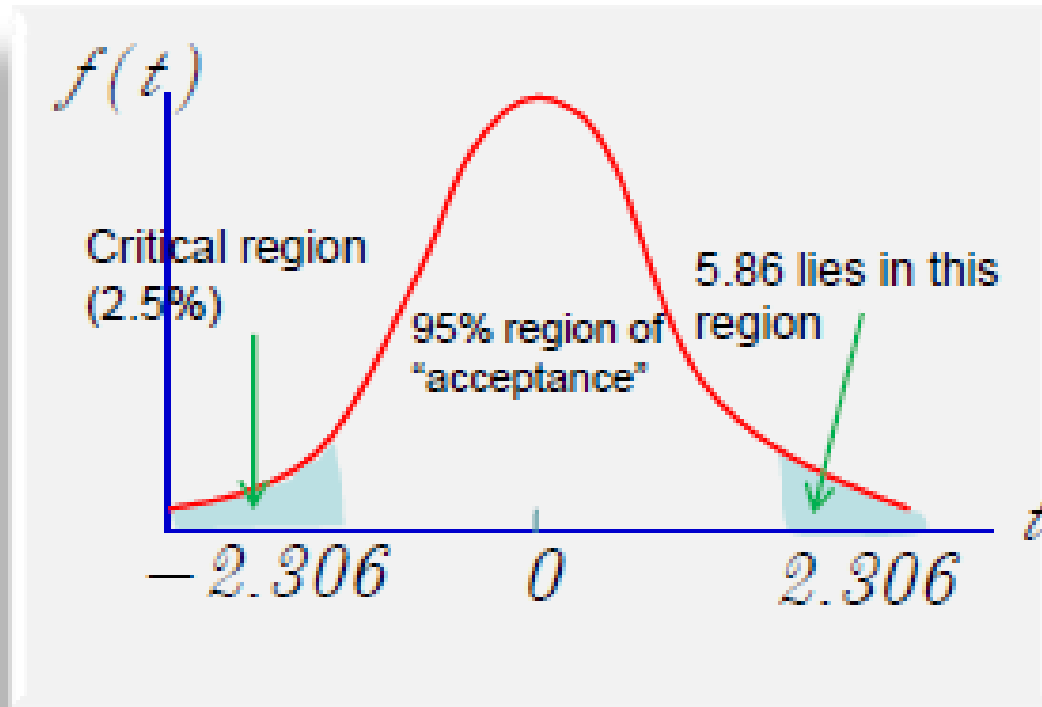
$$H : \beta_1^* \neq 0.3$$

$$\alpha = 0.05$$

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{se(\hat{\beta}_1)} = \frac{0.5091 - 0.3}{0.0357} = 5.86$$

the critical t value at

$$t_{\alpha/2, 8} = t_{0.025, 8} = 2.306$$



Decision:

Since $5.86 > 2.306$, the calculated t value falls in the critical region, we reject the null hypothesis

Example 2

Assume the following population regression equation:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

using the sample of 200 observations we received the ff. regression results:

$$Y_i = 4.233 + 0.469X$$

$$(3.26) \quad (0.213)$$

Now the question is: has X any effect on Y?

The we should under take a hypothesis testing as follows

$$H_0: \beta_1 = 0 \quad H_1: \beta_1 \neq 0$$

Thus we need to form the test function as

$$t = \frac{b_1 - \beta_1}{se(b_1)} \sim t_{n-k} df$$

$$test\ value = \frac{0.469 - 0}{0.213} \sim 198\ df = 2.2$$

Then find the critical value and compared with the test value

Thus the critical value at 5% LS = 1.96, which is less than the test value which is evident to reject the null hypothesis and claim that there is a positive r/ship b/n X & Y.

CI approach

Generally, as noted earlier, a CI could be constructed as:

$$\begin{aligned} & Pr[-t_{\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1^*}{s.e.(\hat{\beta}_1)} \leq t_{\alpha/2}] \\ &= Pr[\hat{\beta}_1 - t_{\alpha/2} s.e.(\hat{\beta}_1) \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2} s.e.(\hat{\beta}_1)] \\ &= 1 - \alpha \end{aligned}$$

where β_1^* is the value of β_1 under H_0 (i.e. $H_0 : \beta_1 = \beta_1^*$) & $t_{\alpha/2}$ are critical t-values.

Procedures in brief:

- 1 Recall the estimated coefficients, such as $\hat{\beta}_1 = -0.46$ & $s.e.(\hat{\beta}_1) = 0.08$.
- 2 Set up the hypothesis: e.g.: $H_0 : \beta_1 = -0.5$; $H_1 : \beta_1 \neq -0.5$.
- 3 Construct a $100(1 - \alpha)\%$ CI under H_0 :

$$(\hat{\beta}_1 - t_{\alpha/2} s.e.(\hat{\beta}_1), \hat{\beta}_1 + t_{\alpha/2} s.e.(\hat{\beta}_1))$$

Decision:

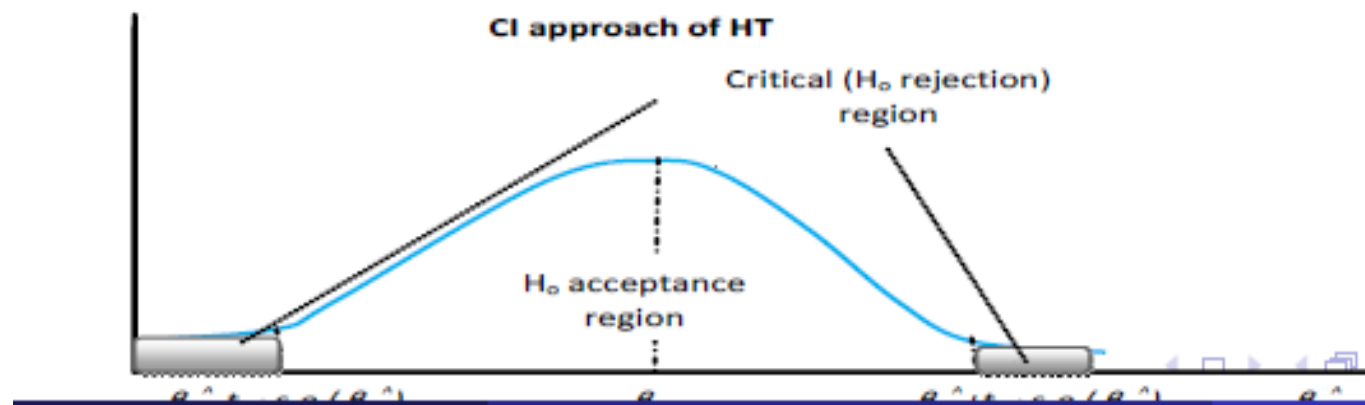
If β_1 under H_0 falls within the CI, DO NOT reject H_0 .
If β_1 under H_0 falls outside the CI, reject H_0 .

Density, $f(\hat{\theta}_1)$

CI approach of HT

Critical (H_0 rejection)
region

H_0 acceptance
region



Prediction

Let the hypothetical data on weekly consumption expenditure(Y) and income (H) of a family is given as

Y, \$	X, \$
70	80
65	100
90	120
95	140
110	160
115	180
120	200
140	220
155	240
150	260

Of the model,

$$Y_i = \beta_1 + \beta_2 X_i + U_i$$

Let the following calculations are obtained, (look Gujarati page 88 for detail)

$$\begin{aligned}\hat{\beta}_1 &= 24.4545 & \text{var}(\hat{\beta}_1) &= 41.1370 & \text{and} & \text{se}(\hat{\beta}_1) &= 6.4138 \\ \hat{\beta}_2 &= 0.5091 & \text{var}(\hat{\beta}_2) &= 0.0013 & \text{and} & \text{se}(\hat{\beta}_2) &= 0.0357 \\ \text{cov}(\hat{\beta}_1, \hat{\beta}_2) &= -0.2172 & \hat{\sigma}^2 &= 42.1591 \\ r^2 &= 0.9621 & r &= 0.9809 & \text{df} &= 8\end{aligned}$$

Hence the estimated model will be

$$\hat{Y}_i = 24.4545 + 0.5091X_i$$

Interpretation

- The value of slope coefficient ($\hat{\beta}_2 = 0.5091$) refers to as **X increases, say, by \$1, the estimated increase in the mean consumption expenditure amounts to about 51 cent.**
- The $r^2 = 0.962$ is showing that about 96% of variation in Y is explained by X.

Thus, predictions can be:

- i. Mean prediction, and
- ii. Individual

Mean prediction

To fix the ideas, assume that $X_0 = 100$ and we want to predict $E(Y | X_0 = 100)$

$$\begin{aligned}\hat{Y}_0 &= \hat{\beta}_1 + \hat{\beta}_2 X_0 \\ &= 24.4545 + 0.5091(100) \\ &= 75.3645\end{aligned}$$

Exercise:

Look at Gujarati page (144-146) for individual prediction

CHAPTER 3: MULTIPLE REGRESSION ANALYSIS

Introduction

- In this chapter least squares regression analysis is generalized to cover the case in which there are several or many explanatory variables in the regression model, rather than just one.
- Here, we will consider regression analysis with one dependent variable(Y) and more than one independent variables (X_s).
The previous SLR is not usually realistic.
- Multiple regression analysis is an extension of simple regression analysis to cover cases in which the dependent variable is hypothesized to depend on more than one explanatory variable.

For example:

- i. The number of children is not affected solely by educational achievement, but also by the wealth, age, biological & other household characteristics.
- ii. Consumption is also affected by many other variables than income

- **The general model we consider in MLR is:**

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + u_i$$

Where, β_0 is still the intercept

β_1 to β_k all called slope parameters

u is still the error term (or disturbance)

- **A Model with Two Explanatory Variables:**

For the sake of simplicity let's take two explanatory regression model (called three-variable model) is:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

where, β_i are parameters estimated

u_i is the disturbance term

- The **disturbance term** is, of similar nature to that in simple regression, reflecting:

- vagueness of theory
- unavailability of data
- core variables vs. peripheral variables
- poor proxy variables
- the basic random nature of human responses
- errors of aggregation
- errors of measurement
- wrong functional form
- and any other (minor) factors, other than x_i that might influence Y

Assumptions of MLR Model

1. Randomness of the error term: The variable u is a real random variable.
2. Zero mean of the error term: $E(u_i) = 0$
3. Homoscedasticity: The variance of each u_i is the same for all the x_i values. i.e. $E(u_i^2) = \sigma_u^2$ (constant)
4. Normality of u
5. No auto or serial correlation: The values of u_i (corresponding to X_i) are independent from the values of any other u_j (corresponding to X_j) for $i \neq j$.
i.e. $E(u_i u_j) = 0$ for $x_i \neq x_j$
6. Independence of u_i and X_i : Every disturbance term u_i is independent of the explanatory variables. i.e. $E(u_i X_{1i}) = E(u_i X_{2i}) = 0$
7. No perfect multicollinearity: The explanatory variables are not perfectly linearly correlated

The Model:

- Given the PRF as: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + U_i$
 $E(Y_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i}$, Since $E(U_i) = 0$
- The sample counter part of the model (SRF) is given as:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$$

Where $\hat{\beta}_j$ are estimates of the β_j and $\hat{Y}$ is known as the predicted value of Y .
Estimation.

Given sample observation on Y , X_1 & X_2 , we estimate our model using the method of least square (OLS).

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + e_i$$

$$e_i = Y_i - \hat{Y} = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}$$

OLS, as usual, minimizes the RSS (residual sum of squares):

$$RSS = \sum \hat{u}_i^2 = \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i})^2$$

Differentiate *RSS* w.r.t. $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$ as follows:

$$(3.1) \quad \frac{\partial RSS}{\partial \hat{\beta}_0} = -2 \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$$

$$(3.2) \quad \frac{\partial RSS}{\partial \hat{\beta}_1} = -2 X_{1i} \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$$

$$(3.3) \quad \frac{\partial RSS}{\partial \hat{\beta}_2} = -2 X_{2i} \sum (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \hat{\beta}_2 X_{2i}) = 0$$

These equations are called **normal equations**.

By rearranging we can get

$$\sum Y = n\hat{\beta}_0 + \hat{\beta}_1 \sum X_{1i} + \hat{\beta}_2 \sum X_{2i} \dots\dots\dots 1)$$

$$\sum X_{1i} Y_i = \hat{\beta}_0 \sum X_{1i} + \hat{\beta}_1 \sum X_{1i}^2 + \hat{\beta}_2 \sum X_{1i} X_{2i} \dots\dots\dots 2)$$

$$\sum X_{2i} Y_i = \hat{\beta}_0 \sum X_{2i} + \hat{\beta}_1 \sum X_{1i} X_{2i} + \hat{\beta}_2 \sum X_{2i}^2 \dots\dots\dots 3)$$

Thus,

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 \text{ ----- (3.7)}$$

Substituting (3.7) in (3.5), we get:

$$\begin{aligned} \sum X_{1i} Y_i &= (\bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2) \sum X_{1i} + \hat{\beta}_1 \sum X_{1i}^2 + \hat{\beta}_2 \sum X_{2i} \\ \Rightarrow \sum X_{1i} Y_i - \hat{Y} \sum X_{1i} &= \hat{\beta}_1 (\sum X_{1i}^2 - \bar{X}_1 \sum X_{2i}) + \hat{\beta}_2 (\sum X_{1i} X_{2i} - \bar{X}_2 \sum X_{2i}) \\ \Rightarrow \sum X_{1i} Y_i - n \bar{Y} \bar{X}_1 &= \hat{\beta}_1 (\sum X_{1i}^2 - n \bar{X}_1^2) + \hat{\beta}_2 (\sum X_{1i} X_{2i} - n \bar{X}_1 \bar{X}_2) \text{ -----} \end{aligned}$$

Now let

$$\begin{aligned} S_{YY} &= \sum Y_i^2 - n \bar{Y}^2 = \sum y_i^2 & S_{11} &= \sum X_{1i}^2 - n \bar{X}_1^2 = \sum d_{1i}^2 \\ S_{1Y} &= \sum X_{1i} Y_i - n \bar{X}_1 \bar{Y} = \sum d_{1i} y_i & S_{22} &= \sum X_{2i}^2 - n \bar{X}_2^2 = \sum d_{2i}^2 \\ S_{2Y} &= \sum X_{2i} Y_i - n \bar{X}_2 \bar{Y} = \sum d_{2i} y_i & S_{12} &= \sum X_{1i} X_{2i} - n \bar{X}_1 \bar{X}_2 = \sum d_{1i} d_{2i} \end{aligned}$$

So, (3.8) becomes

$$S_{1Y} = \hat{\beta}_1 (S_{11}) + \hat{\beta}_2 (S_{12}) \text{ (3.9)}$$

Similarly (3.5) we

$$S_{2Y} = \hat{\beta}_1 (S_{12}) + \hat{\beta}_2 (S_{22}) \text{ (3.10)}$$

Finally, solving eq (3.9) & (3.10) simultaneously gives the following OLS estimators:

$$\begin{aligned}\hat{\beta}_1 &= \frac{S_{22} * S_{1Y} - S_{12} * S_{2Y}}{S_{11} * S_{22} - S_{12}^2} \\ \hat{\beta}_2 &= \frac{S_{11} * S_{2Y} - S_{12} * S_{1Y}}{S_{11} * S_{22} - S_{12}^2} \\ \hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2\end{aligned}$$

Which means:

$$\hat{\beta}_1 = \frac{\Sigma x_1 y \cdot \Sigma x_2^2 - \Sigma x_1 x_2 \cdot \Sigma x_2 y}{\Sigma x_1^2 \cdot \Sigma x_2^2 - \Sigma (x_1 x_2)^2}$$

$$\hat{\beta}_2 = \frac{\Sigma x_2 y \cdot \Sigma x_1^2 - \Sigma x_1 x_2 \cdot \Sigma x_1 y}{\Sigma x_1^2 \cdot \Sigma x_2^2 - \Sigma (x_1 x_2)^2}$$

Coefficient of Determination (R^2)

- The R^2 follows the same formula and same concept as SLR.

Given $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$

R^2 measures the proportion of the variation in Y explained by X_1 & X_2 jointly.

The coefficient of determination is:

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum e_i^2}{\sum y_i^2} \text{-----1)}$$

We know that

$$\underbrace{\sum y_i^2}_{\text{Total sum of square (Total variation)}} = \underbrace{\hat{\beta}_1 \sum x_{1i} y_i + \hat{\beta}_2 \sum x_{2i} y_i}_{\text{Explained sum of square (Explained variation)}} + \underbrace{\sum e_i^2}_{\text{Residual sum of squares (unexplained variation)}}$$

$$\therefore R^2 = \frac{ESS}{TSS} = \frac{\hat{\beta}_1 \Sigma x_{1i} y_i + \hat{\beta}_2 \Sigma x_{2i} y_i}{\Sigma y^2}$$

The square root of R^2 is called the coefficient of multiple correlation (R) and measures the degree of association b/n Y and all the explanatory variables jointly.

Adjusted R^2 ($\bar{R}^2$)

- One difficulty with R^2 is that it can be made large by adding more and more variables, even if the variables added have no economic justification.
- It always favours the model with large k .
- The manipulation of model just to **obtain a high R^2 is not wise.**
- An alternative measure of goodness of fit, called the adjusted R^2 is computed as:

$$\bar{R}^2 = 1 - \frac{\Sigma e_i^2 / n - k}{\Sigma y^2 / n - 1} = 1 - (1 - R^2) \left(\frac{n - 1}{n - k} \right) \text{-----13)}$$

This measure does not always goes up when a variable is added.

- While solving one problem, it losses its interpretation; adjusted R^2 is no longer the percent of variation explained.

N.B.

- For $k > 0$, $\bar{R}^2 < R^2$
 - (As k increases, $\bar{R}^2$ increases less than R^2 .)
- $\bar{R}^2$ can be negative, but R^2 is always non-negative.

Correlation coefficient & partial correlations

Three correlation coefficients: r_{12} (correlation b/n Y & X_2), r_{13} , and r_{23} ; notice that we are letting the subscript 1 represent Y for notational convenience.

$r_{12.3}$ = partial correlation coefficient between Y and X_2 , holding X_3 constant

$r_{13.2}$ = partial correlation coefficient between Y and X_3 , holding X_2 constant

$r_{23.1}$ = partial correlation coefficient between X_2 and X_3 , holding Y constant

For greater precision estimation of the parameters of the 3-var model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

it is good to know the coefficient of correlation b/n X_1 & X_2 ($=r_{12}$).

Generally, the coeff of correlation b/n two random variables is the ratio of their covariance and the product of their standard deviations.

$$r_{12} = \frac{\text{cov}(X_1, X_2)}{s.d(X_1) * s.d(X_2)} = \frac{\sum d_{1i} d_{2i}}{\sqrt{\sum d_{1i}^2 \sum d_{2i}^2}}$$

where $d_{1i} = X_{1i} - \bar{X}_1$ and $d_{2i} = X_{2i} - \bar{X}_2$

So that $r_{12}^2 = \frac{(\sum d_{1i} d_{2i})^2}{\sum d_{1i}^2 \sum d_{2i}^2} = \frac{S_{12}^2}{S_{11} * S_{22}} \dots (3.14)$

Partial r^2 measures the proportion of the variance in the explained variable that only one of the explanatory variable explains.

For instance, given $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$
we may be interested to measure partial r^2 s:

r_{y1}^2 = partial r^2 after the effect of X_2 is removed;

r_{y2}^2 = partial r^2 after the effect of X_1 is removed;

They are computed according to:

$$r_{y1}^2 = \frac{t_1^2}{t_1^2 + (n-3)} \quad \& \quad r_{y2}^2 = \frac{t_2^2}{t_2^2 + (n-3)}$$

$$\text{where } t_1 = \frac{\hat{\beta}_1}{s.e.(\hat{\beta}_1)} \quad \& \quad t_2 = \frac{\hat{\beta}_2}{s.e.(\hat{\beta}_2)}$$

Generally, $r_{yx_i}^2 = \frac{t_i^2}{t_i^2 + (n-k)}; \quad i = 1, 2, \dots, k$

- Let r_{12} is measuring the correlation b/n X_1 & X_2

$$r_{12} = \frac{\text{cov}(X_1, X_2)}{sd(X_1) * sd(X_2)}$$

$$r_{12} = \frac{E[(X_1 - \bar{X}_1)(X_2 - \bar{X}_2)]}{\sqrt{E(X_1 - \bar{X}_1)^2} \sqrt{E(X_2 - \bar{X}_2)^2}}$$

$$r_{12} = \frac{E(x_1 x_2)}{\sqrt{E(x_1)^2} \sqrt{E(x_2)^2}} = \frac{\sum x_1 x_2 / n}{\sqrt{\sum x_1^2 / n} \sqrt{\sum x_2^2 / n}}$$

$$r_{12} = \frac{\sum x_1 x_2}{\sqrt{\sum x_1^2} \sqrt{\sum x_2^2}}$$

$$r_{12}^2 = \frac{(\sum x_1 x_2)^2}{\sum x_1^2 \sum x_2^2} = \frac{s_{12}^2}{s_{11} s_{22}}$$

- Also r^2_{y1}

$$r^2_{Y1} = \frac{t_1^2}{t_1^2 + (n-3)} = \frac{\hat{\beta}_1 / se(\hat{\beta}_1)}{t_1^2 + (n-3)}$$

$$r^2_{Y2} = \frac{t_2^2}{t_2^2 + (n-3)} = \frac{\hat{\beta}_2 / se(\hat{\beta}_2)}{t_2^2 + (n-3)}$$

Generally,

$$r^2_{YXi} = \frac{t_i^2}{t_i^2 + (n-k)}$$

Interpretation

1. Even if $r_{12} = 0$, $r_{12.3}$ will not be zero unless r_{13} or r_{23} or both are zero.
2. If $r_{12} = 0$ and r_{13} and r_{23} are nonzero and are of the same sign, $r_{12.3}$ will be negative, whereas if they are of the opposite signs, it will be positive.

Let Y = crop yield, X_2 = rainfall, and X_3 = temperature. Assume $r_{12} = 0$, that is, no association between crop yield and rainfall.

Assume further that r_{13} is positive and r_{23} is negative. Then, $r_{12.3}$ will be positive; that is, holding temperature constant, there is a positive association between yield and rainfall.

3. The terms $r_{12.3}$ and r_{12} (and similar comparisons) need not have the same sign.
4. In the two-variable case we have seen that r_2 lies between 0 and 1.

Analysis of variance (ANOVA) in regression

ANOVA gives an information about explained and unexplained variation.

The decomposition of the total sample variation in Y used as an alternative for performing a test in regression model.

As a complementary to statistical inference (such as hypothesis testing), let us now try to see regression analysis from ANOVA point of view.

We already know that for simple CLRM $Y_i = \beta_0 + \beta_1 X_i + u_i$:

$$TSS = ESS + RSS \quad \text{And for } d_i = X_i - \bar{X},$$

$$\Rightarrow \sum y_i^2 = \sum \hat{y}_i^2 + \sum \hat{u}_i^2 = \beta_1^2 \sum d_i^2 + \sum \hat{u}_i^2$$

In tabular form, this ANOVA in regression looks like:

Source of variation	Sum of squares(SS)	df	Mean SS(=SS/df)
Due to regression (ESS)	$\sum \hat{y}_i^2 = \beta_1^2 \sum d_i^2$	1	$\beta_1^2 \sum d_i^2$
Due to residuals (RSS)	$\sum \hat{u}_i^2$	$n - 2$	$\frac{\sum \hat{u}_i^2}{n-2} = \hat{\sigma}^2$
Total (TSS)	$\sum y_i^2$	$n - 1$	$\frac{\sum y_i^2}{n-1}$

N.B: As a test statistic it follows the F-function.

With normality assumption, and $H_0 : \beta_1 = 0$, the variable in simple CLRM:

$$F = \frac{MSS \text{ of ESS}}{MSS \text{ of RSS}} = \frac{\hat{\beta}_1^2 \sum x_i^2 / 1}{\sum \hat{u}_i^2 / (n-2)} = \frac{\hat{\beta}_1^2 \sum d_i^2}{\hat{\sigma}^2} = \frac{\hat{\beta}_1}{s.e.(\hat{\beta}_1)} \cdot \frac{\hat{\beta}_1}{s.e.(\hat{\beta}_1)} = t^2$$

follows the F distribution with 1 numerator df and $n - 2$ denominator df.

In simple CLRM, t and F tests provide alternative but complementary ways of testing $H_0 : \beta_1 = 0$ (so that t-test is enough!).

But, in multiple CLRM, F test has several other roles & is a powerful tool of hypothesis testing.

F and R^2 are strongly related:

$$F = \frac{MSS \text{ of ESS}}{MSS \text{ of RSS}} = \frac{ESS/1}{RSS/(n-2)} = \frac{ESS}{TSS-ESS} \cdot (n-2) \\ = \frac{1}{\frac{TSS-ESS}{ESS} \cdot (n-2)} = \frac{1}{(\frac{TSS}{ESS} - 1) \cdot (n-2)} = \frac{1}{(\frac{1}{R^2} - 1) \cdot (n-2)} = \frac{R^2(n-2)}{(1-R^2)}$$

Alternatively, $t^2 = \frac{R^2(n-2)}{(1-R^2)} \iff R^2 = \frac{t^2}{t^2 + (n-2)}$

Variance of estimators: three variable model

Let,

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$$

Where, β_1 is the intercept term. As usual, it gives the mean effect on Y of all the variables excluded from the model. The coefficients β_2 and β_3 are called the **partial regression coefficients**.

The meaning of partial regression coefficient:

$$\beta_2 = \frac{\partial y_i}{\partial x_{2i}}, \text{ holding the value of } X_3 \text{ constant.}$$

Put differently, it gives the “direct” or the “net” effect of a unit change in X_2 on the mean value of Y , net of any effect that X_3 may have on mean Y .

Likewise, β_3 measures the change in the mean value of Y per unit change in X_3 , holding the value of X_2 constant.

-we know that the ff are the **partial regression coefficient**

$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X}_2 - \hat{\beta}_3 \bar{X}_3$$

which is the OLS estimator of the population intercept β_1 .

$$\hat{\beta}_2 = \frac{(\sum y_i x_{2i})(\sum x_{3i}^2) - (\sum y_i x_{3i})(\sum x_{2i} x_{3i})}{(\sum x_{2i}^2)(\sum x_{3i}^2) - (\sum x_{2i} x_{3i})^2}$$

$$\hat{\beta}_3 = \frac{(\sum y_i x_{3i})(\sum x_{2i}^2) - (\sum y_i x_{2i})(\sum x_{2i} x_{3i})}{(\sum x_{2i}^2)(\sum x_{3i}^2) - (\sum x_{2i} x_{3i})^2}$$

where lowercase letters denote deviation forms.

It follows that,

$$\begin{aligned} \text{var}(\hat{\beta}_1) &= \left[\frac{1}{n} + \frac{\bar{X}_2^2 \sum x_{3i}^2 + \bar{X}_3^2 \sum x_{2i}^2 - 2\bar{X}_2 \bar{X}_3 \sum x_{2i} x_{3i}}{\sum x_{2i}^2 \sum x_{3i}^2 - (\sum x_{2i} x_{3i})^2} \right] \cdot \sigma^2 \\ \text{se}(\hat{\beta}_1) &= +\sqrt{\text{var}(\hat{\beta}_1)} \\ \text{var}(\hat{\beta}_2) &= \frac{\sum x_{3i}^2}{(\sum x_{2i}^2)(\sum x_{3i}^2) - (\sum x_{2i} x_{3i})^2} \sigma^2 \end{aligned}$$

or, equivalently,

$$\text{var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_{2i}^2 (1 - r_{23}^2)}$$

$$se(\hat{\beta}_2) = +\sqrt{\text{var}(\hat{\beta}_2)}$$

$$\text{var}(\hat{\beta}_3) = \frac{\sum x_{2i}^2}{(\sum x_{2i}^2)(\sum x_{3i}^2) - (\sum x_{2i}x_{3i})^2} \sigma^2$$

or, equivalently,

$$\text{var}(\hat{\beta}_3) = \frac{\sigma^2}{\sum x_{3i}^2 (1 - r_{23}^2)}$$

$$se(\hat{\beta}_3) = +\sqrt{\text{var}(\hat{\beta}_3)}$$

$$\text{cov}(\hat{\beta}_2, \hat{\beta}_3) = \frac{-r_{23}\sigma^2}{(1 - r_{23}^2)\sqrt{\sum x_{2i}^2}\sqrt{\sum x_{3i}^2}}$$

Hypothesis Testing: Multiple variables case

The Joint Hypothesis Test

Assume the following regression:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + u_i$$

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0$$

H_1 : at least one of the β_k is non-zero

A test of such a hypothesis is called a test of overall significance of the observed or estimated regression line, that is, whether Y is linearly related to $X_1, X_2, \dots, X_k$.

Whenever working with test of several parameters simultaneously we can't use the standard t-test, but using F-test –follows F-distribution.

Let,

Model according to H_0 : $Y_i = \beta_0 + \beta_4 x_{4i} + u_i$

$$H_1: Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + u_i$$

A way to compare these models is to see how different their RSS are from each other.

Thus, the two specifications generate:

- Restricted RSS (RSS_R), and
- An Unrestricted RSS (RSS_U)

In practice these RSS are derived from the separate regression.

Then the test value can be

$$F = \frac{(RSSR - RSSU)/df1}{RSSU/df2} \sim F(\alpha, df1, df2)$$

Where, df1 and df2 are degrees of freedom for numerator and denominator respectively.
 $df1 = (n - k1) - (n - k2) = k2 - k1$

Example

Assume that we have a sample of 1000 observations.

If $RSSR = 17632$ & $RSSU = 9324$, then

$$F = \frac{(17632 - 9324)/2}{9324/(1000 - 4)} = 443.73$$

The critical F value at 5% SL is 4.61

Decision: reject H_0 . That is parameters involved in the test have simultaneous effect on the dependent variable.

Example 1

Let the data on a sample of 5 persons randomly drawn from a large firm giving their annual salaries, years of education, and years of experience.

Where, Y= annual salary; X1= years of educ ; and X2= years of experience

Y	X1	X2	Y- $\hat{Y}$	X1-X1 mean	X2-X2 mean
30	4	10	0	-1	0
20	3	8	-10	-2	-2
26	6	11	6	1	1
24	4	9	-6	-1	-1
40	8	12	10	3	2

Where, y mean=30; x1mean= 5; and x2 mean =10

$$s_{11}=16 \quad s_{12}= 12 \quad s_{1y}= 62 \quad s_{22}=10 \quad s_{2y}= 52 \quad s_{yy}= 272$$

Thus,

$$\hat{\beta}_1 = -0.25 \quad \hat{\beta}_2 = 5.5$$

$$\hat{\alpha} = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 = 30 - (-1.25) - 55 = -23.75$$

$$R^2 = \frac{\hat{\beta}_1 S_{1y} + \hat{\beta}_2 S_{2y}}{S_{yy}} = \frac{271.5}{272} = 0.998$$

Thus the regression equation is

$$\hat{Y} = -23.75 - 0.25X_1 + 5.5X_2 \quad R^2 = 0.998$$

Interpretation:

From the regression result experience is more important.

One more year of experience, by holding education constant, results in an annual increase in salary of 5500 birr. That is if we consider the person with the same level of education, the one with one more year of experience can be expected to have a higher salary of 5500.

Similarly if we consider two people with the same experience, the one with an education of one more year can be expected to have a lower annual salary of 250 birr

EXAMPLE 2: MLR

Let the model is

$$CM_i = \beta_1 + \beta_2 PGNP_i + \beta_3 FLR_i + u_i$$

where, *CM*-child mortality, *PGNP*-per capita GNP, *FLR*-female literacy rate

$$\widehat{CM}_i = 263.6416 - 0.0056 PGNP_i - 2.2316 FLR_i$$

se = (11.5932) (0.0019) (0.2099) $R^2 = 0.7077$
 $\bar{R}^2 = 0.6981^*$

N.B: You can find the raw data on book of “Gujarati- page 180, 6th edition”

INTERPRETATION:

$\beta_2 = 0.0056$ tells us that with the influence of FLR held constant, as PGNP increases, say, by a dollar, on average, child mortality goes down by 0.0056 units.

The coefficient -2.2316 tells us that holding the influence of PGNP constant, on average, the number of deaths of children under age 5 goes down by about 2.23 per thousand live births as the female literacy rate increases by one percentage point.

The intercept value of about 263 means if the values of PGNP and FLR rate were fixed at zero, the mean child mortality rate would be about 263 deaths per thousand live births.

The $R^2 = 0.71$ implies about 71% of the variation in child mortality is explained by PGNP & FLR, a fairly high value considering that the maximum value of R^2 can at most be 1.

Example 3

The following table provides data on:

Y_i = Monthly salary of CEO of business i (in thousands of birr)

X_{1i} = Return on equity of business i (in %)

X_{2i} = Return on bond of business i (in %)

Y	8	3	9	8	7	10	5	10	7	6	5	9	4
X_{1i}	2	1	3	3	2	3	1.5	3	2	0.5	1.5	2.5	1
X_{2i}	1	0.25	1	1.5	0.5	2	0.25	1.5	1	1.5	0.5	1.5	0.5

Suppose we want to fit the following model to the data:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

Based on the above data & model, attempt the following questions:

Determine the estimated OLS regression equation.

Construct the ANOVA table.

Find R^2 and $\bar{R}^2$.

Find the coeff of corr b/n X_1 & X_2 . (r_{12}).

Compute the s.e. of $\hat{\beta}$ s & covariances of all pairs of $\hat{\beta}$ s.

Calculate partial r^2 s i.e. r_{Y1}^2 & r_{Y2}^2 .

Construct a 95% CI for all of the true β s.

Test the following hypotheses at 1% level of significance:

- (i)
$$\left. \begin{array}{l} H_0 : \beta_1 = 0 \\ H_1 : \beta_1 \neq 0 \end{array} \right\} \text{ (using test-of-sign. approach)}$$
- (ii)
$$\left. \begin{array}{l} H_0 : \beta_2 = 3 \\ H_1 : \beta_2 \neq 3 \end{array} \right\} \text{ (using CI approach)}$$
- (iii)
$$\left. \begin{array}{l} H_0 : \beta_1 = \beta_2 = 0 \\ H_1 : \beta_1 \neq 0; \beta_2 \neq 0 \end{array} \right\} \text{ (using F test)}$$

Solution

- We can get that:

$$\begin{aligned} n &= 13 & S_{11} &= 9 & S_{1Y} &= 20.5 \\ k &= 2 & S_{22} &= 3.875 & S_{2Y} &= 12.5 \\ \bar{X}_1 &= 2 & S_{12} &= 3.375 & S_{YY} &= 62 \\ \bar{X}_2 &= 1 \\ \bar{Y} &= 7 \end{aligned}$$

- So that we get:

$$\hat{\beta}_1 = \frac{S_{22} * S_{1Y} - S_{12} * S_{2Y}}{S_{11} * S_{22} - S_{12}^2} = \frac{3.875 * 20.5 - 3.375 * 12.5}{9 * 3.875 - (3.375)^2} \approx 1.59.$$

$$\hat{\beta}_2 = \frac{S_{11} * S_{2Y} - S_{12} * S_{1Y}}{S_{11} * S_{22} - S_{12}^2} = \frac{9 * 12.5 - 3.375 * 20.5}{9 * 3.875 - (3.375)^2} \approx 1.84.$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}_1 - \hat{\beta}_2 \bar{X}_2 = 7 - 1.59 * 2 - 1.84 * 1 = 1.98.$$

- 1. Hence, the estimated OLS regression equation becomes:

$$\hat{Y}_i = 1.98 + 1.59X_{1i} + 1.84X_{2i}$$

(2) $TSS = S_{YY} = 62;$

$$ESS = \hat{\beta}_1 S_{1Y} + \hat{\beta}_2 S_{2Y} = 1.59 * 20.5 + 1.84 * 12.5 = 55.6;$$

$$RSS = TSS - ESS = 62 - 55.6 = 6.4.$$

The ANOVA table of the regression may thus be:

Source of variation	SS	df	MSS
ESS	55.6	2	27.8
RSS	6.4	10	0.64
TSS	62	12	

(3) $R^2 = \frac{ESS}{TSS} = \frac{55.6}{62} \approx 0.90$ $\bar{R}^2 = 1 - (1 - 0.90) \frac{(13-1)}{(13-2)} = 0.88.$

Returns from shares and bonds jointly explain about 88% of the variation in the salary of firms' CEOs.

We notice that $F = \frac{MSS \text{ of } ESS}{MSS \text{ of } RSS} = \frac{27.8}{0.64} = 43.44.$

(4) $r_{12} = \frac{\sum d_{1i} d_{2i}}{\sqrt{\sum d_{1i}^2 * \sum d_{2i}^2}} = \frac{S_{12}}{\sqrt{S_{11} * S_{22}}} = \frac{3.375}{\sqrt{9 * 3.875}} \approx 0.57.$

(5) s.e. & covariances:

$$\hat{\sigma}^2 = \frac{RSS}{n-3} = \frac{6.4}{10} = 0.64.$$

$$s.e.(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{11}(1-r_{12}^2)}} = \sqrt{\frac{0.64}{9(1-0.57^2)}} = 0.32.$$

$$s.e.(\hat{\beta}_2) = \sqrt{\frac{\hat{\sigma}^2}{S_{22}(1-r_{12}^2)}} = \sqrt{\frac{0.64}{3.387(1-0.57^2)}} = 0.53.$$

$$\hat{cov}(\hat{\beta}_1, \hat{\beta}_2) = -\frac{\hat{\sigma}^2 r_{12}}{S_{12}(1-r_{12}^2)} = -\frac{0.64*0.57^2}{3.375(1-0.57^2)} = -0.09.$$

$$\begin{aligned} s.e.(\hat{\beta}_0) &= \sqrt{\frac{\hat{\sigma}^2}{n} + \bar{X}_1^2 \hat{var}(\hat{\beta}_1) + \bar{X}_2^2 \hat{var}(\hat{\beta}_2) + 2\bar{X}_1\bar{X}_2 \hat{cov}(\hat{\beta}_1, \hat{\beta}_2)} \\ &= \sqrt{\frac{0.64}{13} + 2^2 * 0.32^2 + 1^2 * 0.53^2 + 2 * 2 * 1 * -0.09} \\ &= 0.62. \end{aligned}$$

$$\hat{cov}(\hat{\beta}_0, \hat{\beta}_1) = -\left[\bar{X}_1 \hat{var}(\hat{\beta}_1) + \bar{X}_2 \hat{cov}(\hat{\beta}_1, \hat{\beta}_2)\right]$$

$$\begin{aligned} \textcircled{1} \quad \hat{cov}(\hat{\beta}_0, \hat{\beta}_2) &= -\left[\bar{X}_2 \hat{var}(\hat{\beta}_2) + \bar{X}_1 \hat{cov}(\hat{\beta}_1, \hat{\beta}_2)\right] \\ &= -(2 * 0.32 + 1 * -0.09) = -0.55. \\ &= -(1 * 0.53 + 2 * -0.09) = -0.35. \end{aligned}$$

(6) Partial r_s^2 :

$$t_1 = \frac{\hat{\beta}_1}{s.e.(\hat{\beta}_1)} = \frac{1.59}{0.32} = 4.97.$$

$$r_{y1}^2 = \frac{t_1^2}{t_1^2 + (n-3)} = \frac{4.97^2}{4.97^2 + (13-3)} = 0.71.$$

$$t_2 = \frac{\hat{\beta}_2}{s.e.(\hat{\beta}_2)} = \frac{1.84}{0.53} = 3.47.$$

$$r_{y2}^2 = \frac{t_2^2}{t_2^2 + (n-3)} = \frac{3.47^2}{3.47^2 + (13-3)} = 0.55.$$

(7) The 95% CIs (given $\alpha = 0.05$; $t_{\alpha/2, n-k} = t_{0.025, 10} = 2.228$):

$$\begin{aligned} 95\% \text{ CI for } \beta_1 &= \hat{\beta}_1 \pm t_{\alpha/2} * s.e.(\hat{\beta}_1) \\ &= 1.59 \pm 2.228 * 0.32 = (0.88, 2.30). \end{aligned}$$

$$\begin{aligned} 95\% \text{ CI for } \beta_2 &= \hat{\beta}_2 \pm t_{\alpha/2} * s.e.(\hat{\beta}_2) \\ &= 1.84 \pm 2.228 * 0.53 = (0.66, 3.02). \end{aligned}$$

$$\begin{aligned} 95\% \text{ CI for } \beta_0 &= \hat{\beta}_0 \pm t_{\alpha/2} * s.e.(\hat{\beta}_0) \\ &= 1.98 \pm 2.228 * 0.62 = (0.60, 3.36). \end{aligned}$$

(8) Hypothesis testing (given $\alpha = 0.01$;

$$t_{cri} = t_{\alpha/2, n-k} = t_{0.005, 10} = 3.169; F_{cri}^{99\%} = F_{k, (n-k)}^{99\%} = F_{3, 10}^{99\%} = 6.55):$$

(i) $H_0 : \beta_1 = 0$; $H_1 : \beta_1 \neq 0$ (using test-of-sign. approach)

$$\text{We already know that } t_{cal} = t_1 = \frac{\hat{\beta}_1}{s.e.(\hat{\beta}_1)} = \frac{1.59}{0.32} = 4.97.$$

The $H_0 : \beta_1 = 0$ is rejected at 1% since $t_{cal} > t_{cri}$.

(ii) $H_0 : \beta_2 = 3$; $H_1 : \beta_2 \neq 3$ (using CI approach)

$$\begin{aligned} 99\% \text{ CI for } \beta_2 &= \hat{\beta}_2 \pm t_{\alpha/2, (n-k)} * s.e.(\hat{\beta}_2) \\ &= 1.84 \pm 3.169 * 0.53 = (0.16, 3.52). \end{aligned}$$

The $H_0 : \beta_2 = 3$ is NOT rejected at 1% since $\beta_2 = 3$ falls in this interval.

(iii) $H_0 : \beta_1 = \beta_2 = 0$; $H_1 : \beta_1 \neq 0$; $\beta_2 \neq 0$ (using F test)

We already computed that

$$F_{cal} = \frac{ESS/k}{RSS/(n-k)} = \frac{MSS \text{ of } ESS}{MSS \text{ of } RSS} = \frac{27.8}{0.64} = 43.44.$$

Since $F_{cal} > F_{cri}^{99\%} (= 7.56)$, we reject the null that the variables are jointly insignificant (i.e $H_0 : \beta_1 = \beta_2 = 0$).

Exercise

Suppose a hypothetical dataset is given as follows

Y	X_2	X_3
1	1	2
3	2	1
8	3	-3

Based on the data

- Estimate the model: $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$
- Interpret the intercept term and the coefficients
- Compute R^2 and adjusted R^2

Exercise 2

Let the production function is specified as

$$Y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \quad u \sim N(0, \delta^2)$$

where, y =output; x_1 =labor input; and x_2 = capital

From 10 observations the ff result is obtained

$$\begin{aligned} X_1 \text{ mean} &= 10 & x_2 \text{ mean} &= 5 & y \text{ mean} &= 12 & s_{11} &= 12 & s_{12} &= 8 & s_{22} &= 12 & s_{1y} &= 10 & s_{2y} &= 8 \\ s_{yy} &= 10 \end{aligned}$$

- Fit the regression model
- Find the standard error of each parameter
- Find the 95% CI for the parameters
- Test the hypothesis $\beta_1 = \beta_2 = 0$ at 5% SL

The k-variable regression mode

Recall the k-variable (general) multiple linear PRF:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + u_i$$

For n observations, we can write this as:

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{21} + \dots + \beta_k x_{k1} + u_1$$

$$y_2 = \beta_0 + \beta_1 x_{12} + \beta_2 x_{22} + \dots + \beta_k x_{k2} + u_2$$

$$\dots \quad \dots \quad \dots \quad \dots \quad \dots$$

$$y_n = \beta_0 + \beta_1 x_{1n} + \beta_2 x_{2n} + \dots + \beta_k x_{kn} + u_n$$

This can be expressed in matrix format as:

$$\begin{array}{c} \left[\begin{array}{c} y_1 \\ y_2 \\ \dots \\ y_n \end{array} \right] \\ Y \\ (n \times 1) \end{array} = \begin{array}{c} \left[\begin{array}{ccccc} 1 & x_{11} & x_{21} & \dots & x_{k1} \\ 1 & x_{12} & x_{22} & \dots & x_{k2} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_{1n} & x_{2n} & \dots & x_{kn} \end{array} \right] \\ X \\ (n \times k) \end{array} * \begin{array}{c} \left[\begin{array}{c} \beta_0 \\ \beta_1 \\ \dots \\ \beta_k \end{array} \right] \\ \beta \\ (k \times 1) \end{array} + \begin{array}{c} \left[\begin{array}{c} u_1 \\ u_2 \\ \dots \\ u_n \end{array} \right] \\ u \\ (n \times 1) \end{array}$$

The model can then be expressed as:

$$Y = X' \beta + u.$$

Estimating the above model using OLS means getting:

$$Y = X'\hat{\beta} + \hat{u}.$$

The square of the residuals is minimized:

$$\hat{u}'\hat{u} = (Y' - X'\hat{\beta}')(Y - X'\hat{\beta}) = Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta}$$

Taking the 1st order derivative with respect to $\hat{\beta}$ and setting it to zero:

$$\frac{\partial(\hat{u}'\hat{u})}{\partial(\hat{\beta})} = -2X'Y + 2X'X\hat{\beta} = 0 \implies (X'X)\hat{\beta} = X'Y$$

This gives the vector of OLS estimators as:

$$\hat{\beta} = (X'X)^{-1}X'Y$$

Statistical Properties of the Parameters (Matrix) Approach

- They should satisfy the **BLUE** property.

1. Linearity

We know that: $\hat{\beta} = (X'X)^{-1} X'Y$

Let $C = (X'X)^{-1} X'$

$\Rightarrow \hat{\beta} = CY$ ch is a linear function of Y.

2. *Unbiased ness*

$$\hat{\beta} = (X'X)^{-1} X'Y$$

$$\hat{\beta} = (X'X)^{-1} X'(X\beta + U)$$

$$\hat{\beta} = \beta + (X'X)^{-1} X'U \dots\dots\dots$$

$$\text{since } [(X'X)^{-1} X'X = I]$$

$$\begin{aligned} E(\hat{\beta}) &= E\{\beta + (X'X)^{-1} X'U\} \\ &= E(\beta) + E[(X'X)^{-1} X'U] \\ &= \beta + E(X'X)^{-1} X'E(U) \\ &= \beta, \text{ since } E(U) = 0 \end{aligned}$$

3. Minimum variance

Before showing all the OLS estimators are best (possess the minimum variance property), it is important to derive their variance.

We know that, $\text{var}(\hat{\beta}) = E[(\hat{\beta} - \beta)^2] = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']$

$$E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'] =$$

$$\begin{bmatrix} E(\hat{\beta}_1 - \beta_1)^2 & E[(\hat{\beta}_1 - \beta_1)(\hat{\beta}_2 - \beta_2)] & \dots & E[(\hat{\beta}_1 - \beta_1)(\hat{\beta}_k - \beta_k)] \\ E[(\hat{\beta}_2 - \beta_2)(\hat{\beta}_1 - \beta_1)] & E(\hat{\beta}_2 - \beta_2)^2 & \dots & E[(\hat{\beta}_2 - \beta_2)(\hat{\beta}_k - \beta_k)] \\ \vdots & \vdots & \ddots & \vdots \\ E[(\hat{\beta}_k - \beta_k)(\hat{\beta}_1 - \beta_1)] & E[(\hat{\beta}_k - \beta_k)(\hat{\beta}_2 - \beta_2)] & \dots & E(\hat{\beta}_k - \beta_k)^2 \end{bmatrix}$$

$$= \begin{bmatrix} \text{var}(\hat{\beta}_1) & \text{cov}(\hat{\beta}_1, \hat{\beta}_2) & \dots & \text{cov}(\hat{\beta}_1, \hat{\beta}_k) \\ \text{cov}(\hat{\beta}_2, \hat{\beta}_1) & \text{var}(\hat{\beta}_2) & \dots & \text{cov}(\hat{\beta}_2, \hat{\beta}_k) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(\hat{\beta}_k, \hat{\beta}_1) & \text{cov}(\hat{\beta}_k, \hat{\beta}_2) & \dots & \text{var}(\hat{\beta}_k) \end{bmatrix}$$

The above matrix is a symmetric matrix containing variances along its main diagonal and covariance of the estimators every where else. This matrix is, therefore, called the *Variance-covariance* matrix of least squares estimators of the regression slopes. Thus,

$$\text{var}(\hat{\beta}) = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)']$$

- We know that,

$$\hat{\beta} = \beta + (X'X)^{-1} X'U$$

$$\Rightarrow \hat{\beta} - \beta = (X'X)^{-1} X'U$$

Hence,

$$\text{var}(\hat{\beta}) = E\left[\{(X'X)^{-1} X'U\}\{(X'X)^{-1} X'U\}'\right]$$

$$\text{var}(\hat{\beta}) = E\left[(X'X)^{-1} X'UU'X(X'X)^{-1}\right]$$

$$= (X'X)^{-1} X'E(UU')X(X'X)^{-1}$$

$$= (X'X)^{-1} X'\sigma_u^2 I_n X(X'X)^{-1}$$

$$= \sigma_u^2 (X'X)^{-1} X'X(X'X)^{-1}$$

$$\text{var}(\hat{\beta}) = \sigma_u^2 (X'X)^{-1}$$

Assume that $\hat{\beta}$ is an alternative unbiased and linear estimator of β . Suppose that

$$\hat{\beta} = [(X'X)^{-1} X' + B]Y$$

Where B is (k x n) matrix of known constants.

$$\therefore \hat{\beta} = [(X'X)^{-1} X' + B][X\beta + U]$$

$$\hat{\beta} = (X'X)^{-1} X'(X\beta + U) + B(X\beta + U)$$

$$\begin{aligned} E(\hat{\beta}) &= E[(X'X)^{-1} X'(X\beta + U) + B(X\beta + U)] \\ &= E[(X'X)^{-1} X'X\beta + (X'X)^{-1} X'U + BX\beta + BU] \\ &= \beta + BX\beta, \quad [\text{since } E(U) = 0] \end{aligned}$$

Since our assumption regarding an alternative $\hat{\beta}$ is that it is to be an unbiased estimator of β , therefore, $E(\hat{\beta})$ should be equal to β ; in other words (βXB) should be a null matrix.

$$\begin{aligned}
\text{var}(\hat{\beta}) &= E\left[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'\right] \\
&= E\left[\{(X'X)^{-1}X' + B\}Y - \beta\}\{(X'X)^{-1}X' + B\}Y - \beta\}'\right] \\
&= E\left[\{(X'X)^{-1}X' + B\}(X\beta + U) - \beta\}\{(X'X)^{-1}X' + B\}(X\beta + U) - \beta\}'\right] \\
&= E\left[\{(X'X)^{-1}X'X\beta + (X'X)^{-1}X'U + BX\beta + BU - \beta\}\right. \\
&\quad \left.\{(X'X)^{-1}X'X\beta + (X'X)^{-1}X'U + BX\beta + BU - \beta\}'\right] \\
&= E\left[\{(X'X)^{-1}X'U + BU\}\{(X'X)^{-1}X'U + BU\}'\right] \\
&\quad (\because BX = 0) \\
&= E\left[\{(X'X)^{-1}X'U + BU\}\{U'X(X'X)^{-1} + U'B'\}\right] \\
&= E\left[\{(X'X)^{-1}B\}UU'\{X(X'X)^{-1} + U'B'\}\right] \\
&= [(X'X)^{-1}X' + B]E(UU')\{X(X'X)^{-1} + B'\} \\
&= \sigma_u^2 I_n [(X'X)^{-1}X' + B]\{X(X'X)^{-1} + B'\} \\
&= \sigma_u^2 [(X'X)^{-1}X'X(X'X)^{-1} + BX(X'X)^{-1} + (X'X)^{-1}] \\
&= \sigma_u^2 [(X'X)^{-1}X'X(X'X)^{-1} + BX(X'X)^{-1} + (X'X)^{-1}X'B' + BB'] \\
&= \sigma_u^2 [(X'X)^{-1} + BB'] (\because BX = 0)
\end{aligned}$$

$$\text{var}(\hat{\beta}) = \sigma_u^2 (X'X)^{-1} + \sigma_u^2 BB'$$

Or, in other words, $\text{var}(\hat{\beta})$ is greater than $\text{var}(\hat{\beta})$ by an expression $\sigma_u^2 BB'$ and it proves that $\hat{\beta}$ is the best estimator.

INTERVAL ESTIMATION

Which follows a similar procedure like the SLRM.
(Do it by your self)

Chapter 4:

Violations of the Classical Assumptions

4.1. Introduction

- Inference made based on the results of OLS estimation are valid if the assumptions of the CLRMs hold
- Usually, one or more of these assumptions are violated.
- The seriousness and implications of the violations of the assumptions depend on the assumption which is violated.
- In most cases, econometricians can test for and remedy such problems.
- The most commonly violations are:
 - multicollinearity,
 - hetroskedasticity, and
 - autocorrelation

1. Multicollinearity

Nature of multicollinearity

- *Multicollinearity* meant the existence of a “perfect,” or exact, linear relationship among some or all explanatory variables of a regression model.
- For the k -variable regression involving explanatory variable $X_1, X_2, \dots, X_k$ (where $X_1 = 1$ for all observations to allow for the intercept term), an exact linear relationship is said to exist if the following condition is satisfied:

$$\lambda_1 X_1 + \lambda_2 X_2 + \dots + \lambda_k X_k = 0 \quad (4.1)$$

where $\lambda_1, \lambda_2, \dots, \lambda_k$ are constants such that not all of them are zero simultaneously.

- Today, however, the term multicollinearity is used to include the case where the X variables are **intercorrelated** but **not perfectly** so, as follows:

$$\lambda_1 X_1 + \lambda_2 X_2 + \dots + \lambda_k X_k + v_i = 0 \quad (4.2)$$

where v_i is a stochastic error term.

Imperfect MC is a linear functional r/ship b/n two or more independent variables. It is a strong linear r/ship b/n the explanatory variables. It varies in degree from sample to sample

- **Multicollinearity** (or **intercorrelation**) exists when at least some of the predictor variables are correlated among themselves.
- In observational studies, multicollinearity happens more often than not.
- So, we need to understand the effects of multicollinearity on regression analyses.

If multicollinearity is perfect, then

- the regression coefficients are indeterminate;
- their standard errors are infinite.

If multicollinearity is less than perfect, then

- the regression coefficients are determinate; but
- they possess high standard errors, making accuracy difficult.

Example, consider the following hypothetical data:

X_2	X_3	X^*_3
10	50	52
		75
18	90	97
24	120	129
30	150	152

- It is apparent that $X_{3i} = 5X_{2i}$. Therefore, there is perfect collinearity between X_2 and X_3 since the coefficient of correlation r_{23} is unity. The variable X^*_3 was created from X_3 by simply adding to it the following numbers, which were taken from a table of random numbers: 2, 0, 7, 9, 2. Now there is no longer perfect collinearity between X_2 and X^*_3 . ($X_{3i} = 5X_{2i} + v_i$) However, the two variables are highly correlated because calculations will show that the coefficient of correlation between them is 0.9959.

- Why does the classical linear regression model assume that there is no multicollinearity among the X 's? The reasoning is this:
- *If multicollinearity is perfect*, the regression coefficients of the X variables are indeterminate and their standard errors are infinite.
- *If multicollinearity is less than perfect*, the regression coefficients, although determinate, possess large standard errors which means the coefficients *cannot be estimated* with great precision or accuracy.

- Although BLUE, OLS estimators will have large variances and covariance, making precise estimation difficult.
- Confidence intervals tend to be much wider, leading to the acceptance of the zero null hypothesis. more readily.
- t-ratio of one or more coefficients tends to be statistically insignificant.
- Although insignificant t-ratios, R^2 can be very high.
- OLS estimators and their standard errors can be sensitive to small changes in the data.

Sources of Multicollinearity

1. *The data collection method employed*, for example, sampling over a limited range of the values taken by the regressors in the population.
2. *Constraints on the model or in the population being sampled*. For example, in the regression of electricity consumption on income (X_2) and house size (X_3) (High X_2 always mean high X_3).
3. *Model specification*, for example, *adding polynomial* terms to a regression model, especially when the range of the X variable is small.
4. *An overdetermined model*. This happens when the model has more explanatory variables than the number of observations.

An additional reason for multicollinearity, especially in time series data, may be that the regressors included in the model share a *common trend*, that is, they all increase or decrease over time.

The Detection of MC

How do we decide whether an equation has a severe MC problem?

- ❖ In less clear cut cases detecting it may be more difficult
- ✓ But, high coefficient of correlation among X vars, how high is high?
- ✓ Having High VIF- it is an estimate of how much MC has increased the variance of an estimated coefficient. VIF is a method of detecting MC by looking at the extent to which a given Xvar can be explained by all other X vars in the regression.

Run the auxiliary regressions. If near linear dependency, it will display a small SE, a large R^2 and SS F -value.

Looking at the matrix of correlations b/n variables.

Some times none of X vars in the OLS reg is SS(even with wrong sign), though large R^2

VIF

$$VIF_j = (1 - R^2)^{-1}$$

- VIF shows how the variance of an estimator could be inflated by multicollinearity.
- If $r^2_{12} = 1$, VIF gets infinity & if $r^2_{12} = 0$, $VIF=1$ (no multicollinearity).
- Rule of thumb: If VIF exceeds 10, high collinearity.
- This happens for $R^2_j > 0.9$,
- So large SE will lead to large CIs. Imprecision. Also, we might have t-stats that are totally wrong.

Calculating VIF involves three steps:

1. Run an OLS auxiliary regression one for each X vars
2. Calculate the VIF for estimates of the separate regression, hence we can get the corresponding R_i^2
3. Analyze the degree of MC by using the size of VIF

Remedies for MC

- Do nothing. Some argue that if the model is otherwise OK, just ignore it.
- Increasing number of observations/Collect more data
- Dropping of the collinear but insignificant variables, but giving emphasis to specification bias.
- Increase the variance of explanatory variables.
- Respecify the model
- Transform the functional relationship

Consequences of MC

In cases of near or high multicollinearity, one is likely to encounter the following consequences:

1. In the case of perfect multicollinearity regression *coefficients remain indeterminate and their standard errors are infinite*.
2. Although *BLUE*, the OLS estimators *have large variances and covariances*, making precise estimation difficult.
3. Because of consequence 1, the *confidence intervals tend to be much wider*, leading to the *acceptance* of the “zero null hypothesis” (i.e., the true population coefficient is zero) more readily.
4. Also because of consequence 1, the *t* ratio of one or more coefficients tends to be statistically *insignificant*.
5. Although the *t* ratio of one or more coefficients is statistically insignificant, *R² can be very high*.
6. The *OLS estimators and their standard errors can be sensitive* to small changes in the data. The preceding consequences can be demonstrated as follows.

Example

TABLE 10.3
HYPOTHETICAL DATA ON Y , X_2 , AND X_3

Y	X_2	X_3
1	2	4
2	0	2
3	4	12
4	6	0
5	8	16

TABLE 10.4
HYPOTHETICAL DATA ON Y , X_2 , AND X_3

Y	X_2	X_3
1	2	4
2	0	2
3	4	0
4	6	12
5	8	16

Table 10.3: $\hat{Y}_i = 1.1939 + 0.4463X_{2i} + 0.0030X_{3i}$

$$t = (1.5431) \quad (2.4151) \quad (0.0358)$$

$$R^2 = 0.8101$$

$$r_{23} = 0.5523$$

$$\text{cov}(\hat{\beta}_2, \hat{\beta}_3) = -0.00868$$

$$(0.7737) \quad (0.1848) \quad (0.0851)$$

$$df = 2$$

Table 10.4: $\hat{Y}_i = 1.2108 + 0.4014X_{2i} + 0.0270X_{3i}$

$$(0.7480) \quad (0.2721) \quad (0.1252)$$

$$t = (1.6187) \quad (1.4752) \quad (0.2158)$$

$$R^2 = 0.8143$$

$$r_{23} = 0.8285$$

$$\text{cov}(\hat{\beta}_2, \hat{\beta}_3) = -0.0282$$

$$df = 2$$

Example: A hypothetical data on student consumption

CO_i	Yd_i	LA_i
\$2000	\$2500	\$25000
2300	3000	31000
2800	3500	33000
3800	4000	39000
3500	4500	48000
5000	5000	54000
4500	5500	55000

If we run OLS regression on the data, we can obtain

$$\widehat{CO}_i = -367.83 + 0.5113Yd_i + 0.0427LA_i$$

$(1.0307) \quad (0.0942)$
 $t = 0.496 \quad 0.453$
 $\bar{R}^2 = .835$

Where CO- consumption expenditure

Yd –disposable income

LA – liquid asset

On the other hand, if we had consumption as a function of disposable income alone, then we would have

$$\widehat{CO}_i = -471.43 + 0.9714Yd_i$$

(0.157)
t = 6.187
 $\bar{R}^2 = .861$

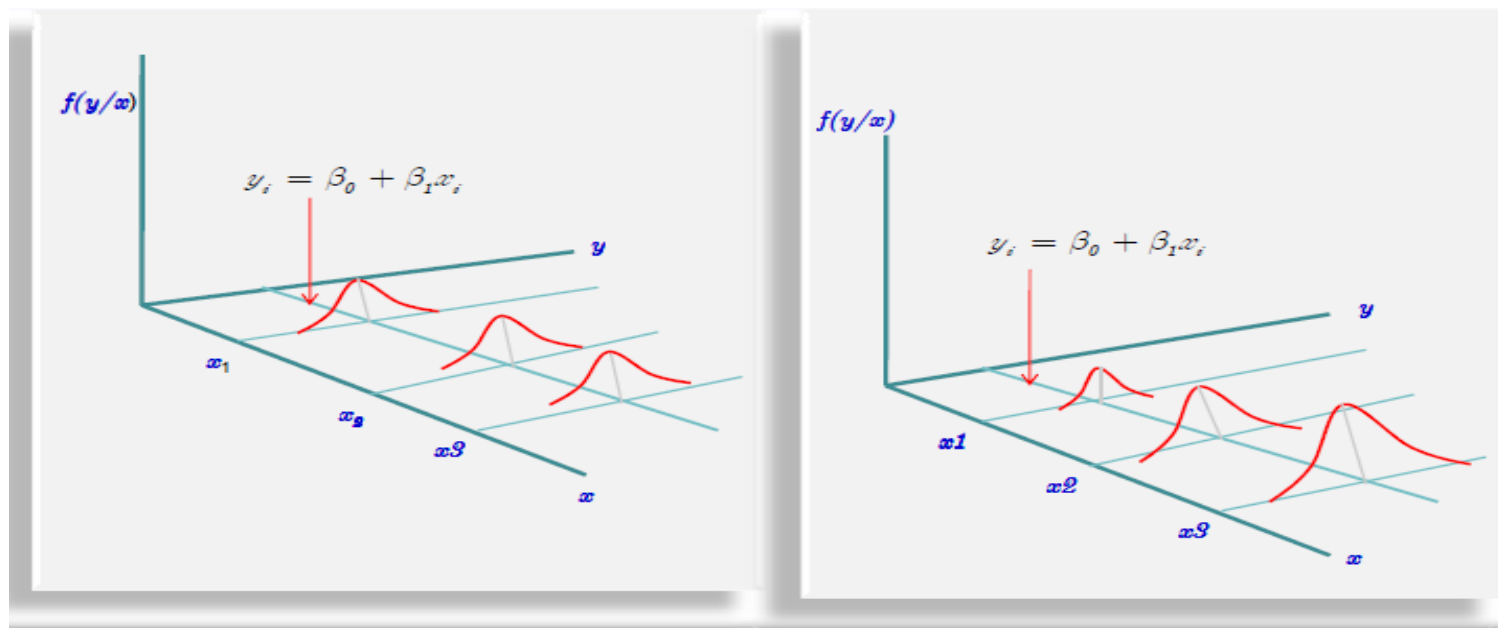
Now the t-score has increased by more than ten fold.

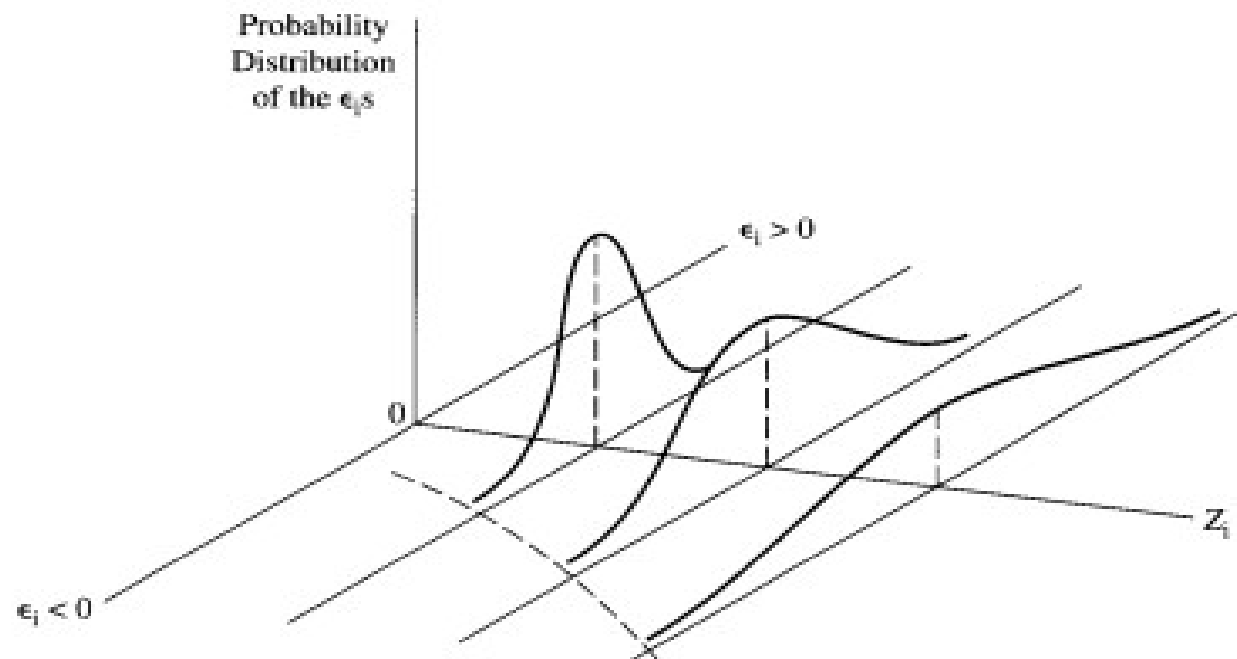
Which equation is better?

Heteroskedasticity

The classical assumption required for OLS estimator to be efficient states that the variance of the error term has to be constant and the same for all observations: **homoskedasticity**. However this assumption is not always realistic.

The distinction between heteroskedasticity and homoskedasticity is important b/c OLS, when applied to heteroskedastic models is no longer the minimum variance estimator. It is still unbiased and consistent however.





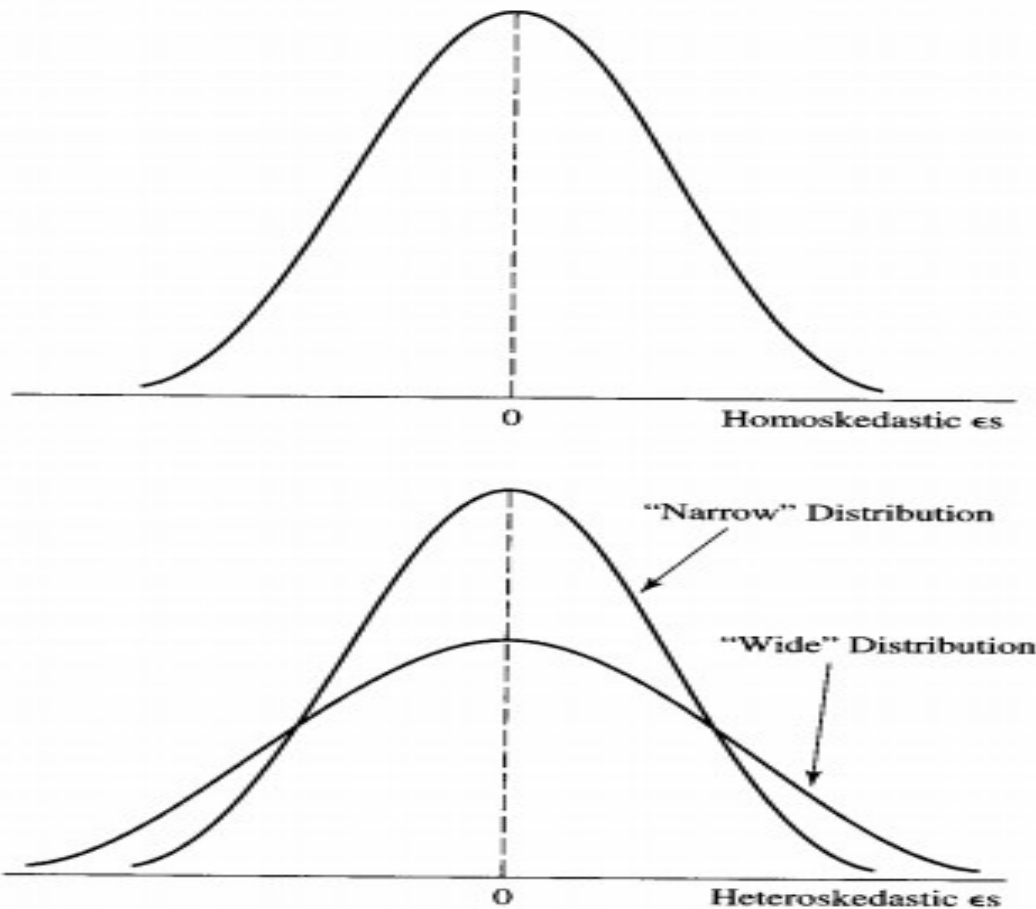
A Heteroskedastic Error Term with Respect to Z_i

If an error term is heteroskedastic with respect to Z_i , the variance of the distribution of the error term changes systematically as a function of Z_i . In this example, the variance is an increasing function of Z_i , as in $\text{VAR}(\epsilon_i) = \sigma^2 Z_i^2$.

Pure Versus Impure Heteroskedasticity

- **Pure heteroskedasticity**: is caused by the error term of a correctly specified equation.
- **Impure heteroskedasticity**: is caused by a specification error such as an omitted variable.
- An omitted variable can cause a heteroskedastic error term because the portion of the omitted effect not represented by one of the included explanatory variables must be absorbed by the error term. If this effect has a heteroskedastic component, the error term of the mis-specified equation might be heteroskedastic even if the error term of the true equation is not.
- Therefore, it is important to be sure that your specification is correct before trying to detect or remedy pure heteroskedasticity.

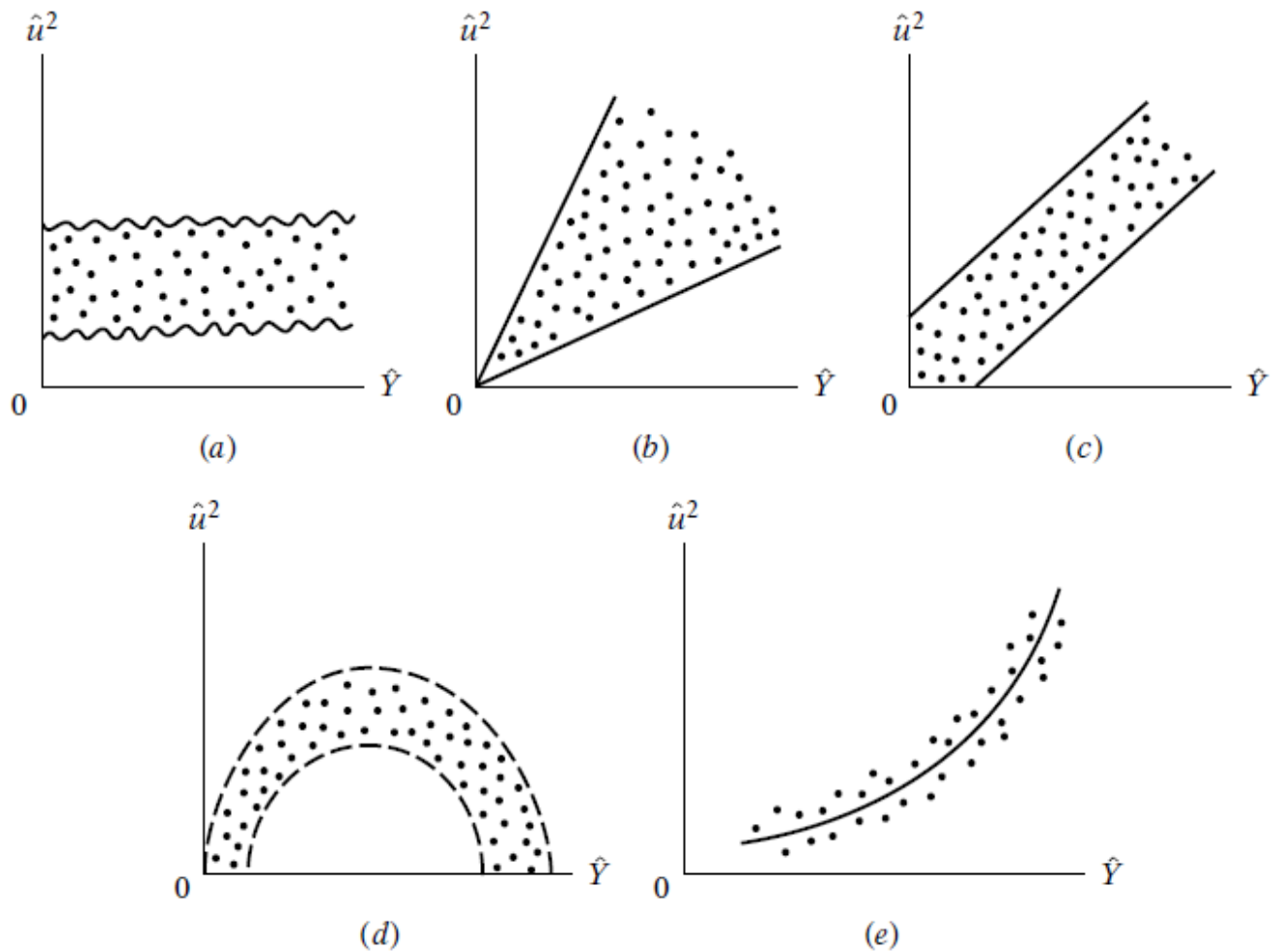
Homoskedasticity Vs Heteroskedasticity



Detecting hetroscedasticity

- Inspection(graphic method)
- It may be generally a rule rather than expectation to test hetroscedasticity in cross sectional data.
- The graph of the **square of residuals against the dependent variable give a rough indication of the existence of hetroscedasticity.**
- If there is a systematic trend in the graph, it may indicate the presence of hetroscedasticity (Figures b, c, d and e)

Graphical method



Statistical tests

- The three most common statistical test procedures to identify a problem of heteroskedasticity are:
 1. Goldfeld-Quant (GQ);
 2. Breusch-Pagan (BP); and
 3. White's tests

GQ test : works under the assumption that the error variance is equal for all observations (i.e. homoskedastic). When this is true the variance of one part of the sample must be the same as the variance of another part of the sample independent on how the sample is sorted.

Procedures:\

1. Sort the sample according to the variable that derives the size of the variance in ascending order, and divide into groups of equal size (into two for small sample)
2. Run the model for each sub sample and compute RSS for each (i.e. $RSS1 =$ & $RSS2 =$)

$$\sum_{i=1}^{n1} e_i^2$$

3. Set the hypothesis: $H_0:$

VS $H_1:$

4. Form the test statistic:

$$F = \frac{S1^2}{S2^2} = \frac{RSS1/(n1-k)}{RSS2/(n2-k)} \sim F(n1-k; n2-k)$$

$$\sum_{i=1}^n e_i^2$$

Example

From the data table given in book of (Gujarati, page 405)

The regression results are:

$$\hat{Y} = 3.4094 + 0.6968X_i$$

$$(8.7049) \quad (0.0744) \quad r^2 = 0.887 \quad RSS1 = 377.17 \quad df=11 \quad \text{for the 1}^{st} \text{ 13 observations}$$

$$\hat{Y} = -28.0272 + 0.7941X_i$$

$$(30.6421) \quad (0.1317) \quad r^2 = 0.7681 \quad RSS1 = 1536.8 \quad df=11 \quad \text{for the last 13 observations}$$

Thus,

$$F = \frac{RSS2/df}{RSS1/df} = \frac{1536.8/11}{377.17/11} = 4.07$$

The critical F value at 5% LS and (11,11)df for numerator and denominator is 2.82.

Conclusion: Reject H_0 (homoskedasticity) implies there is heteroskedasticity in the error variance.

BP test: since it allows for more than one variable at the time to be tested, it is more general

Given the regression equation

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \dots \dots (1)$$

The Breusch-Pagan test for heteroskedasticity

1. Estimate Equation (1) by OLS and obtain the residual $\hat{\varepsilon}$.

2. Run the regression: $\hat{\varepsilon}^2 = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k + e$ and obtain $R_{\hat{\varepsilon}^2}^2$.

3. Form either the F-statistic: $F = \frac{R_{\hat{\varepsilon}^2}^2 / k}{(1 - R_{\hat{\varepsilon}^2}^2) / (n - k - 1)}$

or the LM statistic : $LM = nR_{\hat{\varepsilon}^2}^2$. A small p-value rejects the null of homoskedasticity.

Where R-sqr is the coefficient of determination received from the auxiliary regression of the residual square; and n is number of observation.

➤ The test function follows chi-sqr distribution. The test value should therefore be compared with a critical value received from the chi-sqrb table.

White's Test: it is similar to BP test (large sample test but not depend on any normality assumption)

It is more robust than the previous two methods.

Steps:

1. Estimate the parameters of equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + U$$

and create and save the residual

2. Sqr the residual and run auxiliary regression model as

$$\varepsilon^2 = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_1^2 + \alpha_4 X_2^2 + \alpha_5 X_1 X_2 + U$$

3. Using the result we can calculate the test value of LM. If the test value greater than the critical value chosen, we can reject the **null hypothesis of homoskedasticity**.

Consequences

- ❖ If an error term of an equation is heteroskedastic, there are three major consequences:
 1. Pure heteroskedasticity does not cause bias in the coefficient estimates
 2. Heteroskedasticity typically causes OLS to no longer be the minimum variance estimator (of all the linear unbiased estimators)
 3. Heteroskedasticity causes the OLS estimates of the SE () to be biased, leading to unreliable hypothesis testing i.e **under estimate the variance and the SE** of the coefficients

Pure heteroskedasticity does not cause bias in the coefficient estimates.

This is true because none of the independent variables is correlated with the error term.

Even large positive errors are more likely, so too are large negative errors. The two tend to average each other out, leaving the OLS estimator still unbiased.

As a result we can say that an otherwise correctly specified equation that has pure heteroskedasticity still has the property that:

$$E(\hat{\beta}) = \beta \text{ for all betas}$$

- Lack of bias does not guarantee 'accurate' estimates, especially since heteroskedasticity increases the variance of the estimates, but the distribution of the estimates is still centred around the true beta .
- Equations with impure heteroskedasticity caused by an omitted variable, of course, will have possible specification bias.

Heteroskedasticity typically causes OLS to no longer be the minimum variance estimator (of all the linear unbiased estimators)

- Although the violation of homoskedasticity causes no bias in the coefficient estimates, it results in OLS being no longer of minimum variance. If an error term of an equation is heteroskedastic with respect to a proportionality factor Z :
$$\text{VAR}(\xi_i) = \delta^2 Z_i$$
- This is because the heteroskedastic error term causes the dependent variable to fluctuate, and OLS attributes this fluctuation to the independent variable. This OLS is more likely to misestimate the true β in the face of heteroskedasticity. On balance, the β s are still unbiased because overestimates are just as likely as underestimates.

Heteroskedasticity causes OLS estimates of the $SE(\hat{\beta})$ s to be biased, leading to unreliable hypothesis testing.

- With heteroskedasticity, the OLS formula for the standard error produces biased estimates of the $SE(\hat{\beta})$.
- Because the $SE(\hat{\beta})$ s are used in computing the t-statistic, the t-scores are biased and the hypothesis testing is unreliable.
- Typically heteroskedasticity tends to cause the $SE(\hat{\beta})$ s to be negative so OLS underestimates their size. As the $SE(\hat{\beta})$ s are used in computing the t-statistic, the t-scores are biased (they are usually overestimated – therefore increasing the probability of rejecting a true null hypothesis – Type 1 error) and the hypothesis testing is unreliable.

Remedial measures

- ❖ If heteroskedasticity is found, firstly check for specification errors. If there are no obvious specification errors, the heteroskedasticity is probably pure in nature and one of the following should be considered:
 1. Weighted Least Squares
 2. Heteroskedasticity-Corrected Standard Errors
 3. Redefining the Variables
 4. Using GLS to estimate the unknown population parameter if the cause of heteroskedasticity is known

Weighted Least Squares

- Take an equation with pure heteroskedasticity caused by a proportionality factor Z:
- $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \xi_i$, where the variance of the error term, instead of being constant is $\text{var}(\xi_i) = \delta^2_i = \delta^2 Z_i$

Given that pure heteroskedasticity exists, then we can rewrite the above eqn:

- $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + Z_i U_i$; the error term, $Z_i U_i$ is heteroskedastic b/c $\delta^2 Z_i$, its variance is not constant.

What can be done to make the error term homoskedastic and thereby satisfying the Classical Assumptions?

This general remedy to heteroskedasticity is called (WLS) and is actually a version of GLS.

If δ is known, divide all the variables and the constant by δ

Ex.: Look Gujarati on page 410

The regression results of WLS are as follows:

$$\begin{aligned} \widehat{(Y_i/\sigma_i)} &= 3406.639(1/\sigma_i) + 154.153(X_{1i}/\sigma_i) \\ &\quad (80.983) \quad (16.959) \\ t &= (42.066) \quad (9.090) \\ R^2 &= 0.9993^{33} \end{aligned}$$

For comparison, we give the usual or unweighted OLS regression results:

$$\begin{aligned} \hat{Y}_i &= 3417.833 + 148.767 X_{1i} \\ &\quad (81.136) \quad (14.418) \\ t &= (42.125) \quad (10.318) \quad R^2 = 0.9383 \end{aligned}$$

There are three steps:

1. Dividing the above eqn through the proportionality factor Z, obtaining:

$$Y_i / Z_i = \beta_0 / Z_i + \beta_1 X_{1i} / Z_i + \beta_2 X_{2i} / Z_i + U_i$$

The error term is now U_i , which is homoskedastic

2. Recalculate the data for the transformed variables to conform to this eqn
3. Estimate this eqn with OLS.

Heteroskedasticity – Corrected Standard Errors

- This is the most popular remedy for heteroskedasticity.
- It focuses on improving the estimation of the SE ($\hat{\beta}$)s while still using the OLS estimates of the slope coefficients.
- Since heteroskedasticity causes problems with the SE ($\hat{\beta}$)s but not with the β s, it makes sense to improve the estimation of the SE ($\hat{\beta}$)s in a way that doesn't alter the estimates of the slope coefficients.
- Thus Heteroskedasticity – Corrected (HC) Standard Errors are SE ($\hat{\beta}$)s that have been calculated specifically to avoid the consequences of heteroskedasticity. The HC procedure yields an estimator of the standard errors that, while they are biased, are generally more accurate than uncorrected standard errors for large samples in the face of heteroskedasticity.
- As a result the HC SE ($\hat{\beta}$)s can be used in t-tests and other hypothesis tests in most samples without the errors of inference potentially caused by heteroskedasticity.

- Typically the HC $SE(\hat{\beta})$ s are greater than the OLS $SE(\hat{\beta})$ s – thus lower t-scores and decreasing the probability that a given estimated coefficient will be significantly different from zero.

Problems with HC $SE(\hat{\beta})$ s:

- The technique works best in large samples $\hat{\beta}$
- Not all computer regression packages calculate HC $SE(\hat{\beta})$ s
- Calculation of the HC $SE(\hat{\beta})$ s are beyond the scope of this course

Redefining the Variables

- Redefine the variables in a way that avoids heteroskedasticity. Caution is needed however as redefining involves functional form change that can dramatically change your equation.
- Example, switch from a linear functional form to a double-log functional form. It has inherently less variation than the linear form so its less likely to encounter heteroskedasticity.

Autocorrelation

- It occurs when the classical assumption that $\text{cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } \forall_i \neq j$ is violated
- It occurs most frequently when estimating models using **time series data and hence also called serial correlation.**
- The most popular way of modeling serially correlated disturbances has been to replace the classical assumptions concerning the ε_t by: $\varepsilon_t = \rho\varepsilon_{t-1} + v_t$; $-1 \leq \rho \leq 1$

It is called first order autoregressive (AR) process.

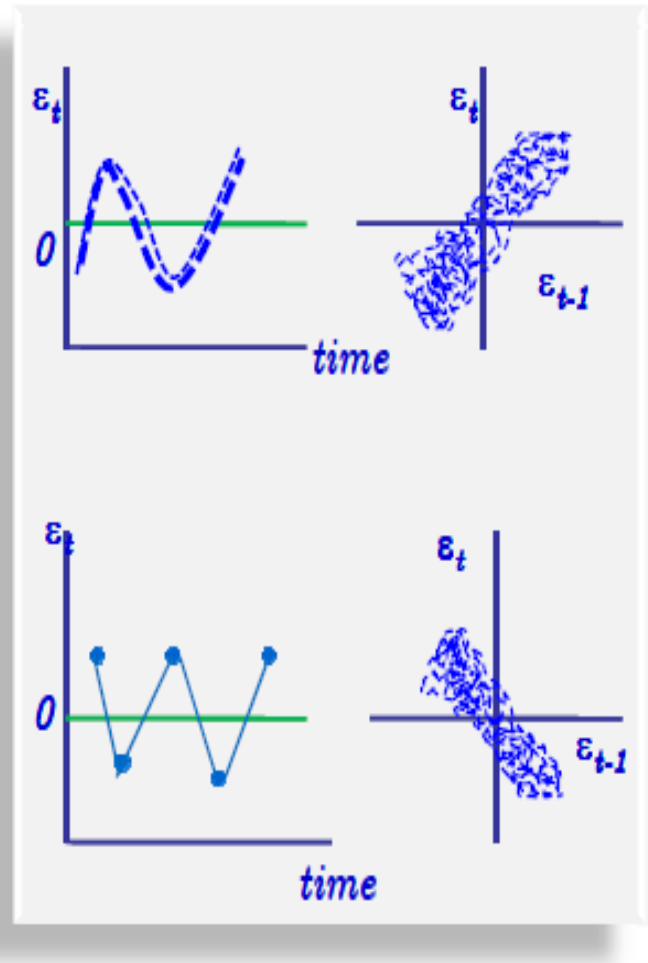
The magnitude of ρ indicates the strength of serial correlation.

If $\rho=0$, no serial correlation. But as $|\rho|=1$, a high degree of serial correlation exists i.e the previous error becomes more important in determining the current value.

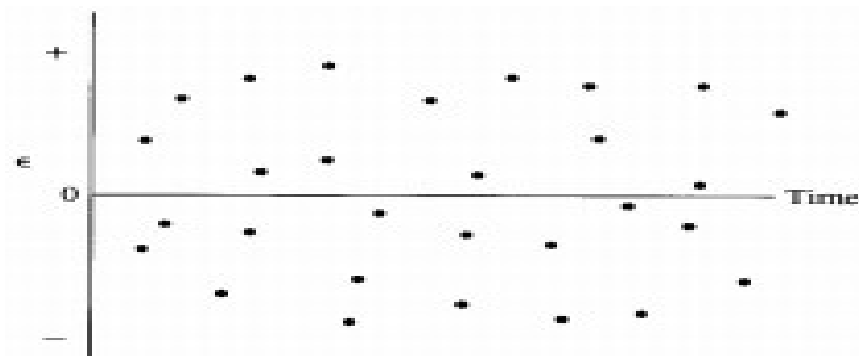
- If $\rho = 0$ there is no correlation
- If $\rho > 0$ there is positive autocorrelation. Positive values of ε_t will be followed by positive values of ε_t or negative values of ε_t will be followed by negative values of ε_t

$$\rho < 0$$

- If $\rho < 0$ there is negative autocorrelation.



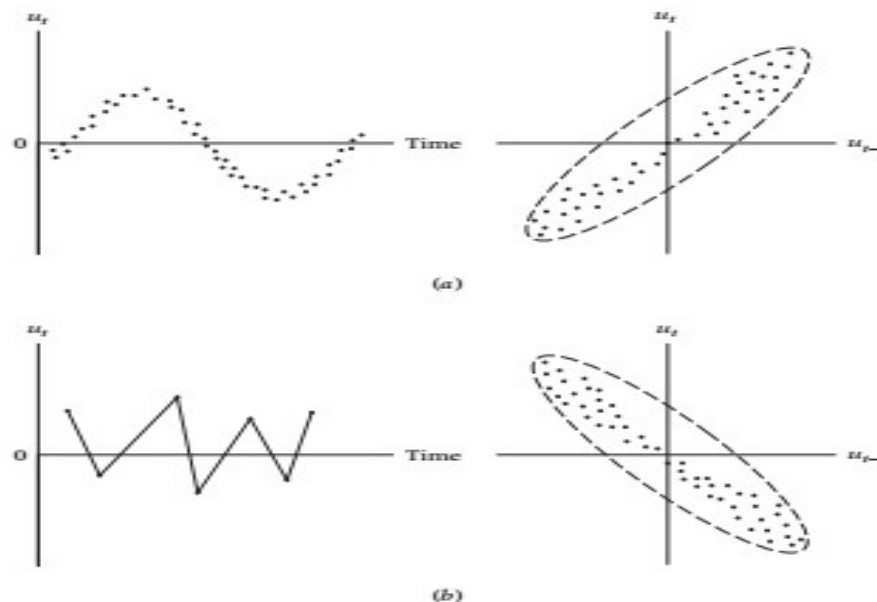
Autocorrelation



No Serial Correlation

With no serial correlation, different observations of the error term are completely uncorrelated with each other. Such error terms would conform to Classical Assumption IV.

(a) Positive and
(b) negative
autocorrelation.



Consequences of serial correlation

There are three major consequences of serial correlation:

Given the model: $Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \epsilon_t$

1. *Pure serial correlation does not cause bias in the coefficient estimates.*

Suppose the error term of the above model has serial correlation as follows

$$\epsilon_t = \rho \epsilon_{t-1} + v_t \text{ where } v_t \text{ is a classical error term.}$$

If the given model has correctly specified and is estimated with OLS, then the estimates of the coefficients of the equation obtained from OLS will be unbiased. i.e.

$$E(\hat{\beta}_1) = \beta_1 \quad \text{and} \quad E(\hat{\beta}_2) = \beta_2$$

Pure serial correlation introduces no bias into the estimation procedure. This conclusion doesn't depend on whether the serial correlation is positive or negative or first order. If the serial correlation is impure, however, bias may be introduced by the use of an incorrect specification.

The consequences of having an autocorrelated error term are very similar to those that appear with a heteroskedastic error term. In short:

1. The estimated slope coefficients are unbiased and consistent
2. With positive autocorrelation the standard errors are unbiased and too small
3. With negative autocorrelation the standard errors are unbiased and too large

Detection of autocorrelation

1. Graphical method: In more advanced time series analysis, graphical methods based on autocorrelation functions and partial autocorrelation functions are used frequently.
2. Other statistic methods:
 - A. Derbin Watson test (DW) test

- It is a test for first order autocorrelation

$$dw = \frac{\sum_{t=2}^n (\varepsilon_t - \varepsilon_{t-1})^2}{\sum_{t=1}^n (\varepsilon_t)^2}$$

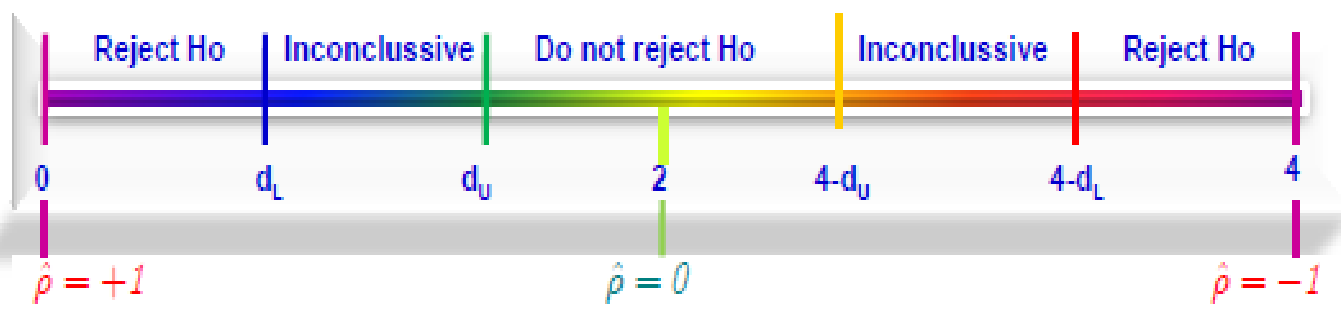
- After some algebra, it can be shown that

$$\hat{\rho} = \frac{\sum \varepsilon_t \varepsilon_{t-1}}{\sum \varepsilon_t^2}$$
$$dw \approx 2(1 - \hat{\rho})$$
$$H_0 : \rho = 0$$

Where ρ is the autocorrelation coefficient

Tests of Autocorrelation: Durbin-Watson(DW) test

- The DW statistic: $dw \approx 2(1 - \hat{\rho})$
is to be read against the Durbin – Watson table under the null $H_0: \rho = 0$.



- Some of the disadvantages of DW test are:
 - It is used only for first – order serial correlation.
 - It is biased towards 2 if there is a lagged dependent variable.
 - It contains inconclusive region.

- DW test statistic only take values b/n 0 & 4 since the autocorrelation coefficient only takes values b/n -1 & 1.
- Since there is no simple distribution function for DW test we cannot establish precise critical value for it.
- Computed d value
 - The closer it is to 0 then we get positive autocorrelation.
 - The closer it is to 4 we get negative autocorrelation.
 - The closer it is to 2 we get no autocorrelation.

Tests of Autocorrelation: LM(Breush-Godfrey) test

- Suppose we estimate the model

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \dots + \beta_n x_{nt} + \varepsilon_t \dots \dots (5)$$

- Assume the error term has the following lag structure

$$\varepsilon_t = \rho_1 \varepsilon_{t-1} + \rho_2 \varepsilon_{t-2} + \rho_3 \varepsilon_{t-3} + \dots + \rho_n \varepsilon_{t-n} + v \dots \dots \dots (6)$$

- If there is no autocorrelation problem, it is naturally expected that the coefficients of the lagged error terms should be zero. Hence, the null is:

$$H_0 : \rho_1 = \rho_2 = \rho_3 = \dots = \rho_n = 0$$

- The procedure is:

1. Estimate the parameters of your main model: $Y_t = \beta_0 + \beta_1 X_t + U_t$
2. Predict the residual $\hat{\varepsilon}_t$
3. Run the auxiliary regression and obtain R sq

$$Y_t = \beta_0 + \beta_1 X_t + \rho U_{t-1} + V_t$$

4. Then, test the null hypothesis $H_0: \rho=0$ using a simple t-test. If you reject it you can conclude that you have autocorrelation.

REMEDIAL MEASURES:

If tests prove that there is a true autocorrelation problem, we may use other techniques such as GLS.

Example:

Assume that we have a time series, and suspect there is auto correlation

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + U_t$$

$$e_t = Y_t - b_0 - b_1 X_t - b_2 X_{2t}$$

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \rho_1 e_{t-1} + \rho_2 e_{t-2} + V_t$$

And receive the following estimates:

$$\hat{Y}_t = 3.241 + 0.378 X_{1t} + 0.562 X_{2t} + 0.324 e_{t-1} + 0.124 e_{t-2}$$

$$\text{s.e} \quad (1.301) \quad (0.091) \quad (0.192) \quad (0.029) \quad (0.101)$$

- By investigating the significance of the coefficients of the two residual terms, we see that the first one is significantly different from zero, while the other is not. thus we conclude that the residual term is serially correlated of the first order only.

Chapter Zero

Dummy Explanatory Variables

- Dummy variables are discrete variables taking a value of '0' or '1'.
- Dummy variables can be used either as explanatory variables or as the dependent variable.

Dummy Explanatory Variables

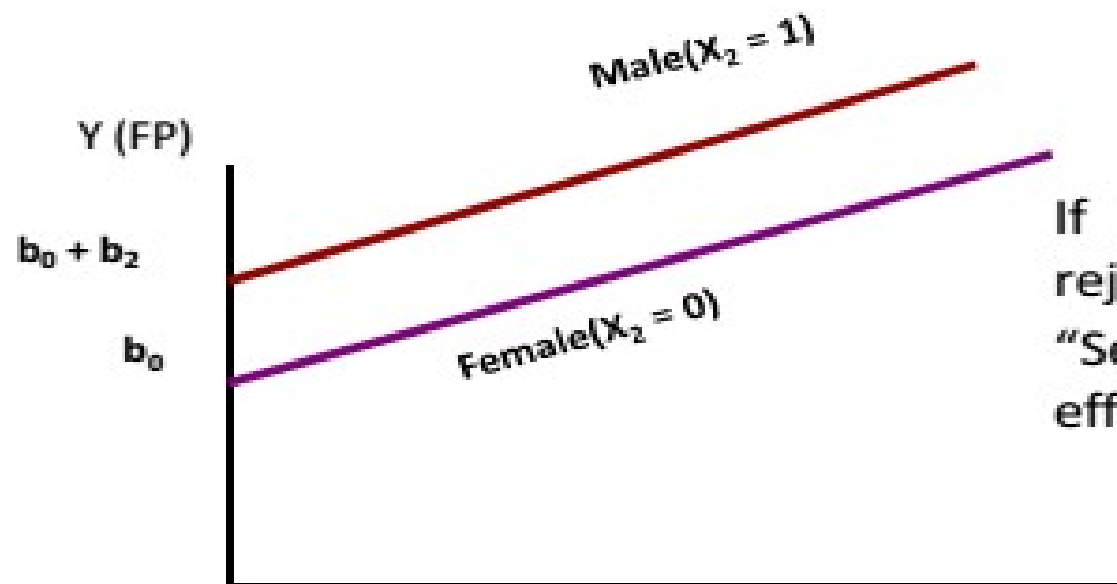
- **Caution in using Dummy Variables**

1. If a qualitative variable has m categories, introduce only $(m-1)$ dummy variables
2. The category with no dummy variable is the benchmark
3. The intercept value represents the measure of the benchmark
4. Coefficients of dummy variables are known as differential intercept coefficients
5. The choice of the benchmark is up to the researcher
6. If we do not introduce an intercept, then we can use as many dummy variables as categories of the qualitative variable

- Different intercepts, same slope

$$\hat{Y} = b_0 + b_1 X_1 + b_2(1) = (b_0 + b_2) + b_1 X_1 \quad \text{Male}$$

$$\hat{Y} = b_0 + b_1 X_1 + b_2(0) = b_0 + b_1 X_1 \quad \text{Female}$$



If $H_0: \beta_2 = 0$ is rejected, then "Sex" has a significant effect on FP

The dummy variable trap

$$Y = \beta_1 + \beta_2 X_2 + \dots + \beta_k X_k + \delta_2 D_2 + \dots \delta_s D_s + u$$

- Suppose that you have a regression model with Y depending on a set of ordinary variables $X_2, \dots, X_k$ and a qualitative variable having s categories.
- We choose one of them as the omitted category (without loss of generality, category 1) and define dummy variables $D_2, \dots, D_s$ for the rest.

The dummy variable trap

$$Y = \beta_1 + \beta_2 X_2 + \dots + \beta_k X_k + \delta_1 D_1 + \delta_2 D_2 + \dots \delta_s D_s + u$$

- What would happen if we did not drop the reference category D_1 ?
- We would fall into the **dummy variable trap**.
- It would be impossible to fit the model as specified.

The dummy variable trap

$$Y = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \delta_1 D_1 + \delta_2 D_2 + \dots \delta_s D_s + u$$

Observation	Category	X_1	D_1	D_2	D_3	D_4
1	4	1	0	0	0	1
2	3	1	0	0	1	0
3	1	1	1	0	0	0
4	2	1	0	1	0	0
5	2	1	0	1	0	0
6	3	1	0	0	1	0
7	1	1	1	0	0	0
8	4	1	0	0	0	1

$$\sum_{i=1}^4 D_i = X_1$$

The dummy variable trap

- Mathematically, we have a special case of exact multicollinearity.
 - If there is no omitted category, there is an exact linear relationship between X_1 and the dummy variables.
- The table above gives an example where there are four categories.
- ❖ If you tried to run the regression anyway, the regression application should detect the problem and do one of two things. It may simply refuse to run the regression.

Alternatively, it may run it, dropping one of the variables in the linear relationship, effectively defining the omitted category.

The dummy variable trap

- ❖ There is another way of avoiding the dummy variable trap. That is to drop the intercept (and X_1). There is no longer a problem because there is no longer an exact linear relationship linking the variables.

Interaction between explanatory variables

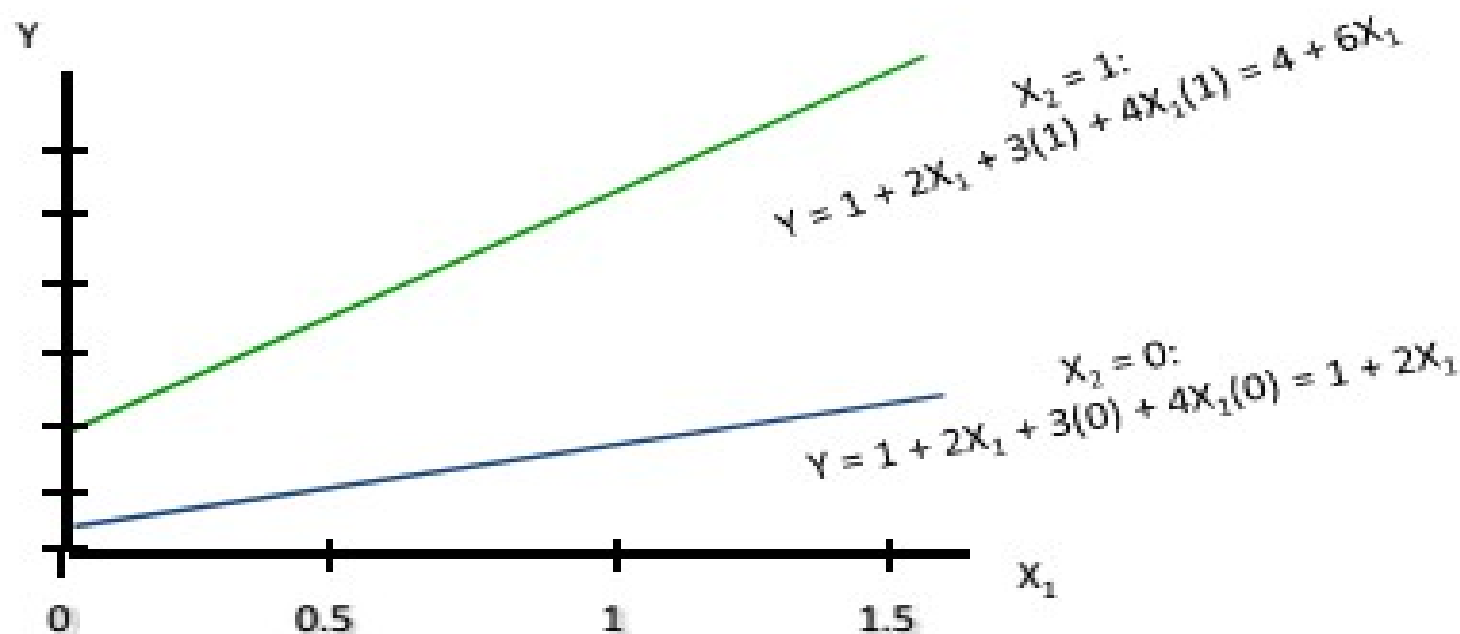
- Hypothesizes interaction between pairs of X variables
 - Response to one X variable may vary at different levels of another X variable
- Contains two-way cross product terms

$$\begin{aligned}\hat{Y} &= b_0 + b_1X_1 + b_2X_2 + b_3X_3 \\ &= b_0 + b_1X_1 + b_2X_2 + b_3(X_1X_2)\end{aligned}$$

- Effect of interaction
 - Without interaction term, effect of X_1 on Y is measured by β_1
 - With interaction term, effect of X_1 on Y is measured by $\beta_1 + \beta_3 X_2$
 - Effect changes as X_2 changes

- Example:** Suppose X_2 is a dummy variable and the estimated regression equation is

$$\hat{y} = 1 + 2X_1 + 3X_2 + 4X_1X_2$$



Slopes are different if the effect of X_1 on Y depends on X_2 value

Thank you!